EasyFlow2 Analysis Workflow User Guide

A static English R Markdown manual and example based on the current Function01-07 modules and EasyFlow analysis script

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Contents

1 Purpose Of This Document

All code chunks in this document are provided as non-executing examples only. Automatic execution is globally disabled with `eval = FALSE`, so knitting to PDF produces a static manual instead of rerunning the workflow.

This document was rewritten directly from the current local EasyFlow codebase:

- `01_Preprocessing.R`
- `02_Interactive_UI.R`
- `03_DimReduction.R`
- `04_Annotation.R`
- `05_MachineLearning.R`
- `06_Visualization.R`
- `07_Stats_and_Export.R`
- `EasyFlow Analysis.R`

The goal of this document is to provide a fully English, publication-friendly R Markdown manual that can be used for:

1. day-to-day execution of the EasyFlow workflow;
2. lab onboarding and internal handoff;
3. project archiving and reproducibility;
4. manuscript methods writing and supplementary documentation.

This version follows the real structure of the current scripts rather than an external helper wrapper, and every adjustable location is annotated with parameter meanings and practical guidance.

2 Codebase Overview

File	Main role	Typical outputs
`01_Preprocessing.R`	FCS import, QC, compensation, transformation, downsampling, panel mapping, gating-template filtering	preprocessed cell-level data frame
`02_Interactive_UI.R`	interactive gating, gating-tree recording, threshold template export	`gated_cells`, gating tree CSV, threshold template CSV, gating tree PDF
`03_DimReduction.R`	clustering, t-SNE, UMAP, benchmark recording	`res_obj`
`04_Annotation.R`	rule-based annotation, negative-marker rule matching, LLM-assisted annotation, label normalization	`final_results`, `Cluster_Annotation_Report.csv`
`05_MachineLearning.R`	large-scale atlas projection and rapid new-sample projection	projected result object, RF model, selected features
`06_Visualization.R`	publication-ready visualization	`ggplot` objects, heatmap object, sample-level PCA
`07_Stats_and_Export.R`	abundance statistics, marker DE, trajectory inference, FCS export, HTML report generation	CSV, PNG, FCS, HTML
`EasyFlow Analysis.R`	end-to-end master script	full analysis workflow template

3 Global Sections That Must Be Edited Before Running

3.1 Project Paths, Function Directory, And Data Directory

The current project-specific path block can be written as follows:

```
project_dir <- "/Volumes/1 TB/lly go colitis facs/EasyFlow/R_20260506"
function_dir <- project_dir
data_dir <- "/Volumes/1 TB/lly go colitis facs/20240619 colitis cytokine/mln"
figure_dir <- file.path(project_dir, "figures")
output_dir <- "/Volumes/Extreme_Pro/Codex/outputs"
gating_tree_pdf <- file.path(project_dir, "00_Gating_Tree_Plot.pdf")
trajectory_dir <- file.path(project_dir, "09_Trajectory_Outputs")

setwd(project_dir)
dir.create(figure_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(output_dir, recursive = TRUE, showWarnings = FALSE)
```

3.1.1 Adjustable parameters in this block

Parameter	Meaning	When to edit
`project_dir`	root working directory for the project	always edit for a new project
`function_dir`	directory containing the `01-07` function scripts	edit if the functions are stored in a subfolder
`data_dir`	directory containing the FCS files	always edit for a new dataset
`figure_dir`	output directory for PDF/PNG figures	edit if you want a different figure output path
`output_dir`	output directory for RDS/CSV/HTML results	edit if you want a different result output path

3.2 Required Packages

Recommended package installation and loading:

```
pkgs <- c(
  "shiny", "plotly", "miniUI", "dplyr", "Rtsne", "ggplot2",
  "stringr", "ggpubr", "scales", "RColorBrewer", "igraph", "RANN",
  "ggridges", "pheatmap", "httr2", "jsonlite", "hexbin", "viridis",
  "flowCore", "uwot", "ranger", "BiocManager", "flowStats",
  "reshape2", "tidyr", "ggalluvial", "ggrepel", "patchwork", "cowplot"
)

new_pkgs <- pkgs[!(pkgs %in% installed.packages()[, "Package"])]
if (length(new_pkgs)) install.packages(new_pkgs)

if (!requireNamespace("FlowSOM", quietly = TRUE)) {
  BiocManager::install("FlowSOM")
}
if (!requireNamespace("slingshot", quietly = TRUE)) {
  BiocManager::install(c("slingshot", "SingleCellExperiment"))
}

invisible(lapply(c(pkgs, "FlowSOM"), library, character.only = TRUE))
```

3.2.1 Adjustable parameters in this block

Parameter	Meaning	When to edit
`pkgs`	list of CRAN dependencies	edit if you add or remove visualization/statistics features

Parameter	Meaning	When to edit
`FlowSOM`	optional dependency for FlowSOM clustering	required when `cluster_method = "flowsom"`
`slingshot`	optional dependency for trajectory inference	required when running `Infer_Trajectory()`
`patchwork` / `cowplot`	required by the marginal-density scatter plot	required when using `Plot_Scatterhist_Line()`

3.3 Automatic Loading Of Core Function Modules

```
r_files <- list.files(function_dir, pattern = "^[0-9]{2}_.*\\.[Rr]$", full.names = TRUE)

message(">>> Loading EasyFlow core modules...")
for (file in sort(r_files)) {
  source(file)
  message("  Loaded: ", basename(file))
}
message(">>> EasyFlow core modules loaded successfully.")
```

3.3.1 Adjustable parameters in this block

Parameter	Meaning	When to edit
`pattern`	determines which R files are loaded	edit if your naming convention changes
`sort(r_files)`	loads modules in prefix order	edit if you want manual control of load order

3.4 Panel Mapping

```
my_panel <- c(
  "[live/dead fv-ef780]-A" = "L/D",
  "BV605-A" = "CD4",
  "BV510-A" = "CD103",
  "FITC-A" = "GM-CSF",
  "PE-Cy7-A" = "IL17a",
  "BV421-A" = "TNFa",
  "PE-A" = "IL6",
  "AF700-A" = "IL10",
  "APC-A" = "IFNg",
  "eFluor450-A" = "Foxp3"
)
```

3.4.1 Adjustable parameters in this block

Parameter	Meaning	When to edit
`my_panel`	mapping from raw cytometer channel names to analysis marker names	must be edited for every experimental panel
left-hand names	actual channel names in the FCS files	must match the acquisition panel exactly
right-hand names	marker names used downstream in analysis	edit when standardizing naming style

3.4.2 Panel mapping recommendations

1. Use a consistent naming style across the project, for example all uppercase human-style names or all mixed-case marker names.
2. Annotation rules, plotting markers, and statistical markers must exactly match the mapped column names.
3. If downstream steps reference `IFNg`, `TNFa`, or `Foxp3`, the mapped column names must match those strings exactly.

3.5 API Key And Proxy Configuration

```
Sys.setenv(OPENROUTER_API_KEY = "")
Sys.setenv(http_proxy = "http://127.0.0.1:7890")
Sys.setenv(https_proxy = "http://127.0.0.1:7890")

tryCatch({
  res <- jsonlite::fromJSON("https://ipapi.co/json/")
  cat("Current R session IP:", res$ip, "\n")
  cat("Current R session region:", res$city, ",", res$country_name, "\n")
}, error = function(e) cat("Network test failed. Proxy or internet access may be unavailable.\n
```

3.5.1 Adjustable parameters in this block

Parameter	Meaning	When to edit
`OPENROUTER_API_KEY`	API key used for LLM-assisted annotation	required when using the LLM path in `Annotate_Clusters_Dual_Track()`
`http_proxy`	HTTP proxy	only set this if your local environment requires a proxy
`https_proxy`	HTTPS proxy	only set this if your local environment requires a proxy

3.5.2 Publication note

For a manuscript-ready or shareable version of this document, keep the API key as a placeholder rather than embedding a real secret in the file.

4 Recommended Centralized Configuration

Although the original master script spreads parameters across multiple steps, long-term maintenance is much easier if the parameters are defined centrally first and then reused downstream. The following block maps directly onto the current `EasyFlow Analysis.R` logic.

```
group_keywords <- c("wt", "ko")

preprocess_cfg <- list(
  n_cells_per_sample = 50000,
  do_QC = TRUE,
  do_compensation = TRUE,
  transform_method = "arcsinh",
  cofactor = 150
)

cluster_cfg <- list(
  k_neighbors = 30,
  cluster_mode = "B",
  do_scale = TRUE,
  cluster_method = "louvain",
  flowsom_k = 20,
  tsne_perplexity = NULL,
  umap_n_neighbors = 15,
  umap_min_dist = 0.2
)

annotation_cfg <- list(
  model = "openai/gpt-5.4",
  use_llm_only = TRUE,
  ontology = c("CD4 T cell", "Unknown"),
  max_clusters_per_batch = 20,
  base_population_context = paste(
    "All cells in this analysis were pre-gated as CD4+ T cells",
    "from mLN.",
    "All annotations must stay within the CD4 T-cell universe."
  ),
  allow_functional_prefix = TRUE,
  unknown_threshold = 35,
  annotation_granularity = "fine"
)
```

```r
projection_cfg <- list(
  batch1_folder = "/Volumes/1 TB/lly go colitis facs/20240619 colitis cytokine/mln",
  batch2_folder = "/Volumes/1 TB/lly go colitis facs/situmulate mlns",
  batch1_label = "Batch1_20240619_mLN",
  batch2_label = "Batch2_stimulated_mLN",
  gating_template_path = "/Volumes/Extreme_Pro/Codex/Gating_Thresholds_Template.csv",
  n_cells_per_sample = 1500000,
  target_node = "cell_live_cell_CD4",
  margin = 0.02,
  target_label = "Cluster_Name",
  cv_folds = 5,
  rejection_threshold = 0.55,
  use_class_weights = TRUE,
  rare_cell_boost = 2.0,
  batch_correction = "adtnorm_lite",
  batch_col = "Batch",
  reference_batch = "Batch1_20240619_mLN",
  batch_diagnostic_pdf = "ADTnorm_Batch_Correction_Diagnostics.pdf"
)

plot_order_cfg <- list(
  Group = c("wt", "ko")
)

plot_cfg <- list(
  my_dim = "UMAP",
  my_color = "Cluster_Name",
  my_split = "Group",
  my_target_marker = "IL17a",
  my_group_var = "Group",
  my_fill_var = "Group",
  my_facet_var = "Cluster_Name",
  my_bar_x = "Group",
  my_bar_fill = "Cluster_Name",
  my_sankey_axis1 = "Group",
  my_sankey_axis2 = "Cluster_Name",
  my_x_marker = "IL17a",
  my_y_marker = "Foxp3",
  my_density_grp = "Cluster_Name",
  my_exp_grp = "Cluster_Name",
  my_dot_markers = NULL,
  my_conf_true = "CellType",
  my_conf_pred = "Cluster_Name",
  my_sample_group = "Group",
  my_sample_id = "SampleID"
)
```

```
stats_cfg <- list(
  my_stat_target = "Cluster_Name",
  my_stat_group = "Group",
  my_stat_method = "wilcoxon",
  my_stat_trans = "clr"
)

report_cfg <- list(
  project_name = "EasyFlow2 Final Report",
  out_file = "EasyFlow2_Final_Report.html",
  template = "manuscript"
)
```

4.0.1 Meaning of each configuration block

Configuration block	Role
`group_keywords`	defines how experimental groups are inferred from filenames
`preprocess_cfg`	controls preprocessing, QC, compensation, transformation, and downsampling
`cluster_cfg`	controls clustering, t-SNE, and UMAP
`annotation_cfg`	controls rule-based annotation, LLM-assisted annotation, and label granularity
`projection_cfg`	controls two-batch large-scale projection, ADTnorm-lite correction, and rejection thresholding
`plot_order_cfg`	fixes plotting order of groups and cell populations
`plot_cfg`	controls the variables used in the figure generation section
`stats_cfg`	controls abundance statistics and transformation method
`report_cfg`	controls HTML report title and template

5 Standard Workflow

5.1 Step 1: Data Reading, QC, And Preprocessing

```
raw_data_rds <- file.path(output_dir, "raw_data.rds")

if (file.exists(raw_data_rds)) {
  raw_data <- readRDS(raw_data_rds)
  message("Loaded raw_data from: ", raw_data_rds)
```

```r
} else {
  raw_data <- Prepare_Raw_Data(
    work_dir = data_dir,
    group_keywords = group_keywords,
    n_cells_per_sample = preprocess_cfg$n_cells_per_sample,
    do_QC = preprocess_cfg$do_QC,
    do_compensation = preprocess_cfg$do_compensation,
    transform_method = preprocess_cfg$transform_method,
    cofactor = preprocess_cfg$cofactor
  )
  raw_data <- Apply_Panel_Mapping(raw_data, my_panel)
  if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)
  saveRDS(raw_data, file = raw_data_rds)
  message("Saved raw_data to: ", raw_data_rds)
}

head(raw_data)
cat(
  "Preprocessing completed:",
  nrow(raw_data), "cells across",
  length(unique(raw_data$SampleID)), "samples.\n"
)
```

5.1.1 Adjustable parameters in `Prepare_Raw_Data()`

Parameter	Current example value	Meaning	Practical guidance
`work_dir`	`data_dir`	directory containing the FCS files	must be changed to your real dataset path
`group_keywords`	`c("wt", "ko")`	labels groups using filename keywords	replace with the group keywords used in your study
`n_cells_per_sample`	`50000`	maximum number of cells retained per sample	`20,000` to `50,000` is common for reference atlas construction
`do_QC`	`TRUE`	enables automated debris and doublet filtering	recommended for routine flow cytometry analyses
`do_compensation`	`TRUE`	applies the embedded compensation matrix from the FCS file	usually recommended for fluorescence flow cytometry
`transform_method`	`"arcsinh"`	transformation method, one of `"arcsinh"`, `"logicle"`, or `"none"`	`arcsinh` is usually the most robust default

Parameter	Current example value	Meaning	Practical guidance
`cofactor`	`150`	ArcSinh cofactor	typically `150` for flow cytometry and `5` for CyTOF

5.1.2 Adjustable parameters in `Apply_Panel_Mapping()`

Parameter	Meaning
`df`	the preprocessed data frame returned from the previous step
`panel_list`	the channel-to-marker mapping table

5.1.3 Typical outputs from this step

- `raw_data`
- `SampleID`
- `Group`
- mapped marker columns based on `my_panel`

5.2 Step 2: Interactive Gating

```
gated_cells_rds <- file.path(output_dir, "gated_cells.rds")
file.exists(file.path(output_dir, "gated_cells.rds"))
if (file.exists(gated_cells_rds)) {
  gated_cells <- readRDS(gated_cells_rds)
  message("Loaded gated_cells from: ", gated_cells_rds)
} else {
  gated_cells <- raw_data
  message("No gated_cells.rds found; using raw_data as fallback.")
}
head(gated_cells)
cat(
  "Gating completed:",
  nrow(gated_cells), "cells retained across",
  length(unique(gated_cells$CellType)), "annotated populations.\n"
)
```

```
if (file.exists(gating_tree_pdf)) {
  knitr::include_graphics(gating_tree_pdf)
} else {
  cat("Gating tree PDF was not found at:", gating_tree_pdf, "\n")
}
```

5.2.1 Parameters for `Run_Universal_Gating()`

Parameter	Meaning
`merged_df`	the preprocessed cell-level data frame

5.2.2 Adjustable parameters in `Plot_Gating_Tree()`

Parameter	Default	Meaning
`csv_path`	`"Gating_Tree_Structure.csv"`	path to the gating-tree structure CSV
`save_path`	`"00_Gating_Tree_Plot.pdf"`	output path for the gating-tree PDF

5.2.3 Typical output files from the gating step

File	Meaning
`Gating_Thresholds_Template.csv`	1% to 99% threshold template for each gated node across channels
`Gating_Tree_Structure.csv`	gating parent-child structure with cumulative counts
`00_Gating_Tree_Plot.pdf`	visualized gating tree

5.2.4 Adjustable elements during interactive gating

1. `x_axis` and `y_axis` determine which two markers are displayed for manual selection.
2. `Show density` controls whether density contours are overlaid.
3. `Set as global gate` keeps only the currently selected cells.
4. `Remove selection` removes the selected debris or unwanted events.
5. `Annotate selection` assigns a name to the selected subpopulation.

5.3 Step 3: Dimensionality Reduction And Clustering

```
ref_results_rds <- file.path(output_dir, "ref_results.rds")

if (file.exists(ref_results_rds)) {
  ref_results <- readRDS(ref_results_rds)
  message("Loaded ref_results from: ", ref_results_rds)
} else {
  ref_results <- Run_Analysis_On_Gated_Data(
    gated_data = gated_cells,
```

```
    k_neighbors = cluster_cfg$k_neighbors,
    cluster_mode = cluster_cfg$cluster_mode,
    do_scale = cluster_cfg$do_scale,
    cluster_method = cluster_cfg$cluster_method,
    flowsom_k = cluster_cfg$flowsom_k,
    tsne_perplexity = cluster_cfg$tsne_perplexity,
    umap_n_neighbors = cluster_cfg$umap_n_neighbors,
    umap_min_dist = cluster_cfg$umap_min_dist
  )
  if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)
  saveRDS(ref_results, file = ref_results_rds)
  message("Saved ref_results to: ", ref_results_rds)
}

cat(
  "Clustering completed:",
  nrow(ref_results$Data), "cells and",
  length(unique(ref_results$Data$Cluster)), "clusters.\n"
)
```

5.3.1 Adjustable parameters in `Run_Analysis_On_Gated_Data()`

Parameter	Current example value	Meaning	Practical guidance
`gated_data`	`gated_cells`	the gated cell-level data	fixed output from the previous step
`k_neighbors`	`30`	number of neighbors used for graph construction	increase for larger or more complex datasets if needed
`cluster_mode`	`"B"`	`"A"` clusters previously named cells only, `"B"` clusters all gated-in cells	`"B"` is usually preferred for global atlas construction
`do_scale`	`TRUE`	applies Z-score scaling before clustering	recommended when markers are on different scales
`cluster_method`	`"louvain"`	one of `"louvain"`, `"leiden"`, or `"flowsom"`	Louvain or Leiden are usually the safest general choices
`flowsom_k`	`20`	number of FlowSOM meta-clusters	only used when `cluster_method = "flowsom"`
`tsne_perplexity`	`NULL`	t-SNE perplexity	leaving this as `NULL` lets the function choose a safe value automatically

Parameter	Current example value	Meaning	Practical guidance
`umap_n_neighbors`	`15`	number of UMAP neighbors	increase if you want smoother manifolds
`umap_min_dist`	`0.2`	UMAP minimum distance	decrease if you want tighter embeddings

5.3.2 Returned object structure

`ref_results` usually contains:

- `ref_results$Data`: cell-level data, cluster labels, t-SNE, and UMAP coordinates
- `ref_results$Info$UMAP_Model`: reusable UMAP model for downstream projection
- `ref_results$Info$Benchmarks`: runtime benchmarks

5.4 Step 4: Dual-Track AI Annotation

The legacy rule format is still supported:

```
custom_rules <- list(
  "CD4 T cell" = c("CD4"),
  "Treg" = c("CD4", "Foxp3"),
  "Th1" = c("CD4", "IFNg"),
  "Th17" = c("CD4", "IL17a")
)
```

The current annotation module is better used with explicit positive and negative marker logic:

```
custom_rules <- list(
  "CD4 T cell" = list(
    positive = c("CD4"),
    negative = c("CD8")
  ),
  "Treg" = list(
    positive = c("CD4", "Foxp3"),
    negative = c("IFNg", "TNFa")
  ),
  "Th1" = list(
    positive = c("CD4", "IFNg"),
    negative = c("Foxp3")
  ),
  "Th17" = list(
    positive = c("CD4", "IL17a"),
    negative = c("Foxp3")
  )
)
```

5.4.1 Recommended annotation call

```
final_results_rds <- file.path(output_dir, "final_results.rds")

if (file.exists(final_results_rds)) {
  final_results <- readRDS(final_results_rds)
  message("Loaded final_results from: ", final_results_rds)
} else {
  my_ontology <- c(
    "CD4 T cell",
    "Treg",
    "Th1",
    "Th17",
    "Unknown"
  )

  final_results <- Annotate_Clusters_Dual_Track(
    res_obj = ref_results,
    rules = custom_rules,
    model = annotation_cfg$model,
    use_llm_only = annotation_cfg$use_llm_only,
    ontology = annotation_cfg$ontology,
    max_clusters_per_batch = annotation_cfg$max_clusters_per_batch,
    base_population_context = annotation_cfg$base_population_context,
    allow_functional_prefix = annotation_cfg$allow_functional_prefix,
    unknown_threshold = annotation_cfg$unknown_threshold,
    annotation_granularity = annotation_cfg$annotation_granularity
  )
  if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)
  saveRDS(final_results, file = final_results_rds)
  message("Saved final_results to: ", final_results_rds)
}

cat(
  "Annotation completed. Available label columns:",
  paste(
    intersect(
      c("Cluster_Name", "Coarse_Name", "Fine_Name", "Major_Lineage", "Minor_Lineage", "State_Na
      colnames(final_results$Data)
    ),
    collapse = ", "
  ),
  "\n"
)
```

5.4.2 Adjustable parameters in `Annotate_Clusters_Dual_Track()`

Parameter	Current example value	Meaning	Practical guidance
`res_obj`	`ref_results`	clustering result object	fixed output from the previous step
`rules`	`custom_rules`	rule-based annotation table	strongly recommended when canonical marker combinations are known
`model`	`"openai/gpt-5.4"`	LLM used for annotation	adjust according to your available API endpoint
`use_llm_only`	`TRUE`	skips rule-based matching and runs the full LLM path	set to `FALSE` if you want rules first, LLM second
`ontology`	`c("CD4 T cell", "Unknown")`	constrains allowed annotation labels	strongly recommended for publication-grade annotation
`max_clusters_per_batch`	20	number of clusters submitted per LLM batch	reduce if prompts become too large
`base_population_context`	CD4 T-cell context sentence	provides parent-lineage context to the LLM	highly recommended if upstream gating already defined a biological universe
`allow_functional_prefix`	`TRUE`	allows functional-state prefixes in labels	set to `FALSE` for more conservative naming
`unknown_threshold`	35	labels low-confidence clusters as Unknown	lower it to reduce Unknown labels; increase it for stricter annotation
`annotation_granularity`	`"fine"`	one of `"fine"`, `"coarse"`, or `"both"`	`fine` or `coarse` are most common in manuscripts

5.4.3 Manual review after annotation

```
# Run this block manually after reviewing Cluster_Annotation_Report.csv
corrected_report <- read.csv("Cluster_Annotation_Report.csv")
final_results$Data$Cluster_Name <- corrected_report$fine_name[
  match(final_results$Data$Cluster, corrected_report$cluster)
]
# Re-save after manual correction
saveRDS(final_results, file = file.path(output_dir, "final_results.rds"))
```

5.4.4 Typical outputs from this step

- `Cluster_Annotation_Report.csv`
- `final_results$Data$Cluster_Name`
- `final_results$Data$Coarse_Name`
- `final_results$Data$Fine_Name`
- `final_results$Data$Major_Lineage`
- `final_results$Data$Minor_Lineage`
- `final_results$Data$State_Name`

5.5 Step 5: Central Control Panel For Plotting And Reporting Variables

This section mirrors the central variable assignment area in the current master script.

```
my_dim <- plot_cfg$my_dim
my_color <- plot_cfg$my_color
my_split <- plot_cfg$my_split

my_target_marker <- plot_cfg$my_target_marker
my_group_var <- plot_cfg$my_group_var
my_fill_var <- plot_cfg$my_fill_var
my_facet_var <- plot_cfg$my_facet_var

my_bar_x <- plot_cfg$my_bar_x
my_bar_fill <- plot_cfg$my_bar_fill
my_sankey_axis1 <- plot_cfg$my_sankey_axis1
my_sankey_axis2 <- plot_cfg$my_sankey_axis2

my_x_marker <- plot_cfg$my_x_marker
my_y_marker <- plot_cfg$my_y_marker
my_density_grp <- plot_cfg$my_density_grp

my_exp_grp <- plot_cfg$my_exp_grp
my_dot_markers <- plot_cfg$my_dot_markers

my_conf_true <- plot_cfg$my_conf_true
my_conf_pred <- plot_cfg$my_conf_pred

my_sample_group <- plot_cfg$my_sample_group
my_sample_id <- plot_cfg$my_sample_id

final_results <- Set_Plot_Order(
  final_results,
  order_list = plot_order_cfg
)
```

5.5.1 Adjustable parameters in this section

Parameter	Meaning
`my_dim`	chooses `"UMAP"` or `"tSNE"` as the main embedding view
`my_color`	determines how the main embedding is colored, for example `Cluster` or `Cluster_Name`
`my_split`	optional faceting variable, for example `Group`; set to `NULL` for no splitting
`my_target_marker`	marker used in violin, bar, and ridge plots
`my_group_var`	grouping variable for expression comparisons
`my_fill_var`	fill variable for violin-style plots
`my_facet_var`	faceting variable for violin/bar plots
`my_bar_x`	x-axis variable for composition plots
`my_bar_fill`	fill variable for composition plots
`my_sankey_axis1` / `my_sankey_axis2`	axes used in the alluvial plot
`my_x_marker` / `my_y_marker`	two markers used in the dual-marker scatter plot
`my_density_grp`	grouping variable used in the dual-marker scatter plot
`my_exp_grp`	grouping variable for dotplot and heatmap
`my_dot_markers`	optional custom marker list for the dotplot; `NULL` enables automatic detection
`my_conf_true` / `my_conf_pred`	true-label and predicted-label columns for the confusion matrix
`my_sample_group` / `my_sample_id`	color and label variables for sample-level PCA
`plot_order_cfg`	explicit global plotting order, strongly recommended for publication figures

5.6 Step 6: One-Click Publication-Ready Figure Generation

5.6.1 Main embedding figure

```
p_dim <- Plot_Dim_Reduction(
  final_results,
  dim_method = my_dim,
  color_by = my_color,
  split_by = my_split
)

print(p_dim)

ggsave(
  paste0("01_", my_dim, "_by_", my_color,
         ifelse(is.null(my_split), "", paste0("_split_", my_split)),
```

```
        ".pdf"),
  plot = p_dim,
  width = 12,
  height = 6
)
```

5.6.1.1 Adjustable parameters in `Plot_Dim_Reduction()`

Parameter	Default	Meaning
`dim_method`	`"UMAP"` / `"tSNE"`	chooses which 2D embedding to display
`color_by`	`"Cluster"`	variable used for coloring
`split_by`	`NULL`	optional faceting variable

5.6.2 Composition plots

```
p_comp_fill <- Plot_Composition(
  final_results,
  x_var = my_bar_x,
  fill_var = my_bar_fill,
  position = "fill",
  correct_sample_bias = TRUE
)

p_comp_stack <- Plot_Composition(
  final_results,
  x_var = my_bar_x,
  fill_var = my_bar_fill,
  position = "stack",
  correct_sample_bias = TRUE
)
print(p_comp_fill)
```

5.6.2.1 Adjustable parameters in `Plot_Composition()`

Parameter	Default	Meaning
`x_var`	`"Group"`	x-axis grouping variable
`fill_var`	`"Cluster_Name"`	stacked fill variable
`position`	`"fill"` or `"stack"`	proportional or absolute stacked plot
`correct_sample_bias`	`TRUE`	computes proportions per sample before averaging by group

Parameter	Default	Meaning
`sample_col`	`"SampleID"`	sample identifier column

5.6.3 Alluvial plot

```
p_sankey <- Plot_Alluvial(
  final_results,
  axis1 = my_sankey_axis1,
  axis2 = my_sankey_axis2,
  fill_var = my_bar_fill
)

print(p_sankey)
```

5.6.3.1 Adjustable parameters in `Plot_Alluvial()`

Parameter	Default	Meaning
`axis1`	`"Group"`	first axis
`axis2`	`"Cluster_Name"`	second axis
`fill_var`	`"Cluster_Name"`	variable controlling flow colors

5.6.4 Dotplot

```
p_dot <- Plot_Dotplot(
  final_results,
  markers = my_dot_markers,
  group_by = my_exp_grp
)

print(p_dot)
```

5.6.4.1 Adjustable parameters in `Plot_Dotplot()`

Parameter	Default	Meaning
`markers`	`NULL`	custom marker list; `NULL` auto-detects numeric marker columns
`group_by`	`"Cluster_Name"`	grouping variable shown on the y-axis

5.6.5 All-in-one violin plot

The current master script has already switched from a manually written violin block to `Plot_Violin_AllInOne()`. This is one of the main upgrades that should be highlighted in the documentation.

```
p_v <- Plot_Violin_AllInOne(
  res_obj = final_results,
  marker = my_target_marker,
  cluster_by = my_facet_var,
  group_by = my_group_var,
  fill_by = my_fill_var,
  add_boxplot = TRUE,
  trim = FALSE,
  add_stats = TRUE,
  stat_method = "wilcox.test",
  stat_label = "p.signif",
  hide_ns = FALSE,
  p_y_offset = 0.08,
  logfc_y_offset = 0.16,
  logfc_digits = 2
)

print(p_v)
```

5.6.5.1 Adjustable parameters in `Plot_Violin_AllInOne()`

Parameter	Current example value	Meaning
`marker`	controlled by `my_target_marker`	marker whose expression distribution will be shown
`cluster_by`	`"Cluster_Name"`	first-level grouping variable
`group_by`	`"Group"`	second-level grouping variable inside each cluster
`fill_by`	`NULL`, usually set to `group_by`	violin fill variable
`add_boxplot`	`TRUE`	overlays boxplots on violins
`trim`	`FALSE`	trims violin tails or not
`add_stats`	`TRUE`	adds statistical comparison labels
`stat_method`	`"wilcox.test"`	one of `"wilcox.test"` or `"t.test"`
`stat_label`	`"p.signif"`	significance stars or formatted p-values; use `"p.format"` if needed
`comparisons`	`NULL`	usually left at default in the current workflow

Parameter	Current example value	Meaning
`hide_ns`	`FALSE`	hides non-significant labels if set to `TRUE`
`violin_alpha`	`0.7`	violin transparency
`logfc_digits`	`2`	number of decimal places shown in log2FC
`logfc_prefix`	`"log2FC="`	prefix used for the logFC label
`p_y_offset`	`0.08`	relative height of p-value labels
`logfc_y_offset`	`0.16`	relative height of logFC labels
`point_size`	`NULL`	optionally overlays jitter points

5.6.6 LogFC-annotated barplot

```
p_bar <- Compare_Populations_Barplot(
  final_results,
  marker = my_target_marker,
  group_by = my_group_var,
  reference_group = "wt",
  show_error_bars = TRUE,
  logfc_threshold = 0.1,
  p_threshold = 0.05,
  facet_by = my_facet_var
)

p_bar
```

5.6.6.1 Adjustable parameters in `Compare_Populations_Barplot()`

Parameter	Current example value	Meaning
`marker`	`my_target_marker`	marker to compare
`group_by`	`my_group_var`	grouping variable used for comparison
`reference_group`	`"wt"`	reference group for logFC calculation
`show_error_bars`	`TRUE`	displays SEM/error bars
`logfc_threshold`	`0.1`	only labels logFC above this threshold
`p_threshold`	`0.05`	only labels statistically significant comparisons
`facet_by`	`my_facet_var`	optional faceting variable

5.6.7 Heatmap

```
hm_res <- Plot_Heatmap(
  final_results,
  group_by = my_exp_grp,
  scale = "row",
  cluster_rows = TRUE,
  cluster_cols = TRUE
)

if (!is.null(hm_res$Plot$gtable)) {
  grid::grid.newpage()
  grid::grid.draw(hm_res$Plot$gtable)
}
```

5.6.7.1 Adjustable parameters in `Plot_Heatmap()`

Parameter	Default	Meaning
`group_by`	`"Cluster"`	grouping variable used before aggregation
`scale`	`"row"`	row scaling, column scaling, or none
`cluster_rows`	`TRUE`	clusters rows if `TRUE`
`cluster_cols`	`TRUE`	clusters columns if `TRUE`
`color_palette`	`NULL`	custom heatmap palette
`show_border`	`TRUE`	shows tile borders
`fontsize`	`10`	text size

5.6.8 Confusion matrix

```
if (my_conf_true %in% colnames(final_results$Data) &&
    my_conf_pred %in% colnames(final_results$Data)) {
  p_conf <- Plot_Confusion_Matrix(
    final_results,
    true_label = my_conf_true,
    pred_label = my_conf_pred
  )

  p_conf
} else {
  cat("Confusion matrix skipped because the requested columns were not found.\n")
}
```

5.6.8.1 Adjustable parameters in `Plot_Confusion_Matrix()`

Parameter	Default	Meaning
`true_label`	`"CellType"`	ground-truth label column
`pred_label`	`"Cluster_Name"`	predicted label column

5.6.9 Ridge plot

```
p_density <- Plot_Density_Line(
  final_results,
  marker = my_target_marker,
  group_by = my_group_var
)

p_density
```

5.6.9.1 Adjustable parameters in `Plot_Density_Line()`

Parameter	Default	Meaning
`marker`	required	marker used for density estimation
`group_by`	`"Cluster_Name"`	grouping variable
`custom_colors`	`NULL`	custom color palette

5.6.10 Dual-marker scatter plot with marginal densities

```
p_scatterhist <- Plot_Scatterhist_Line(
  final_results,
  x_marker = my_x_marker,
  y_marker = my_y_marker,
  group_by = my_density_grp
)

p_scatterhist
```

5.6.10.1 Adjustable parameters in `Plot_Scatterhist_Line()`

Parameter	Current default	Meaning
`x_marker`	required	x-axis marker
`y_marker`	required	y-axis marker
`group_by`	`"Cluster_Name"`	grouping variable used for colors
`custom_colors`	`NULL`	custom color vector

Parameter	Current default	Meaning
`point_size`	`0.5`	main scatter-point size
`point_alpha`	`0.6`	main scatter-point transparency
`density_linewidth`	`0.6`	marginal density line width
`margin_ratio`	`0.15`	relative size of marginal panels
`density_height`	`0.8`	controls how tall the marginal density curves appear

5.6.11 Sample-level PCA

```
p_sample_pca <- Plot_Sample_PCA(
  final_results,
  color_by = my_sample_group,
  label_by = my_sample_id
)

p_sample_pca
```

5.6.11.1 Adjustable parameters in `Plot_Sample_PCA()`

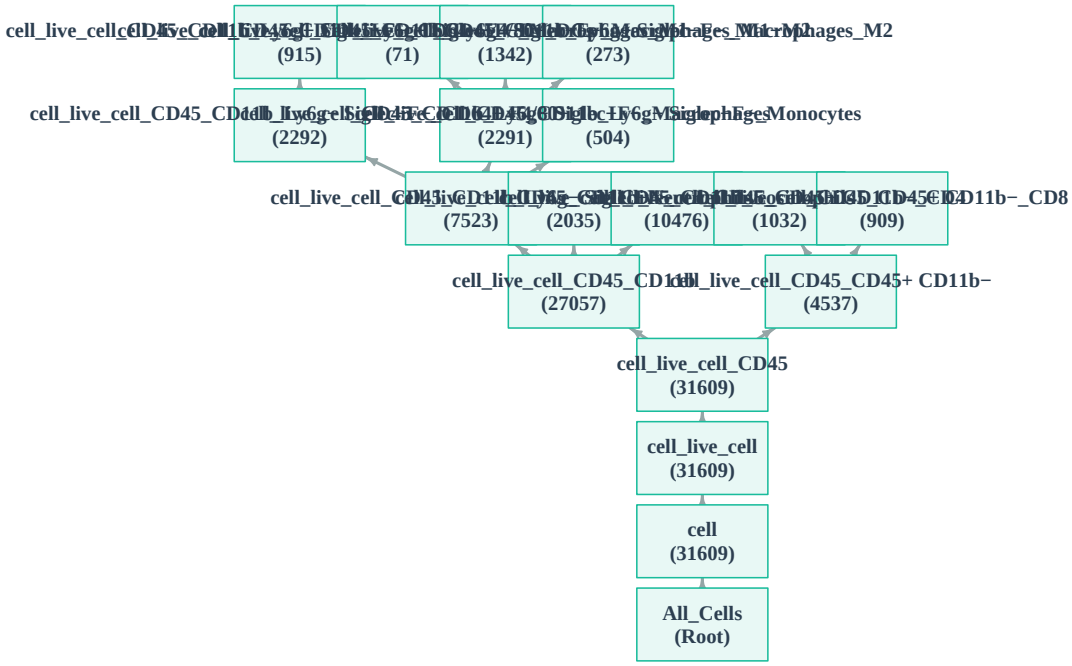
Parameter	Default	Meaning
`color_by`	`"Group"`	grouping variable used for sample coloring
`label_by`	`"SampleID"`	sample label column

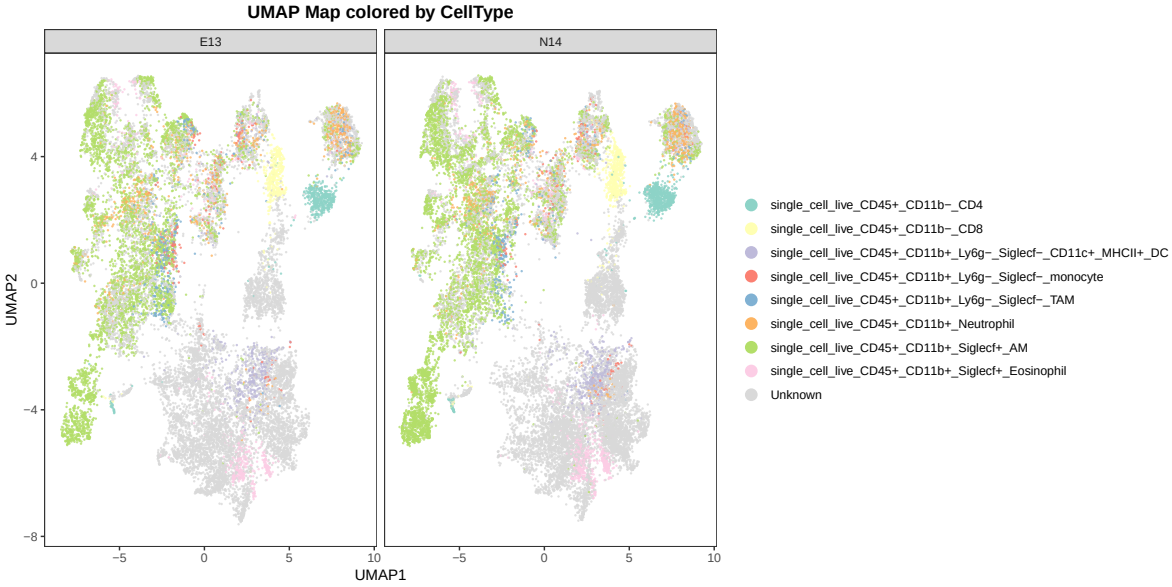
5.7 Embed Pre-Generated Figures In The Static PDF Manual

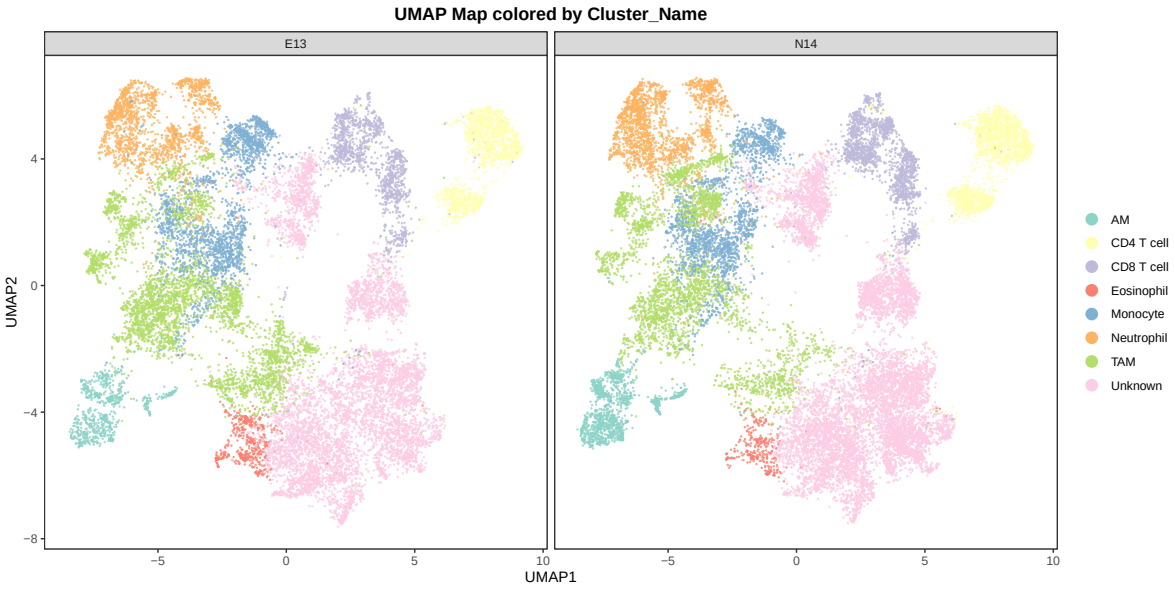
Because this document is configured as a static manual with global `eval = FALSE`, the plotting chunks above are displayed as reproducible example code but are not executed during PDF export. Therefore, those chunks will not automatically create or embed figures.

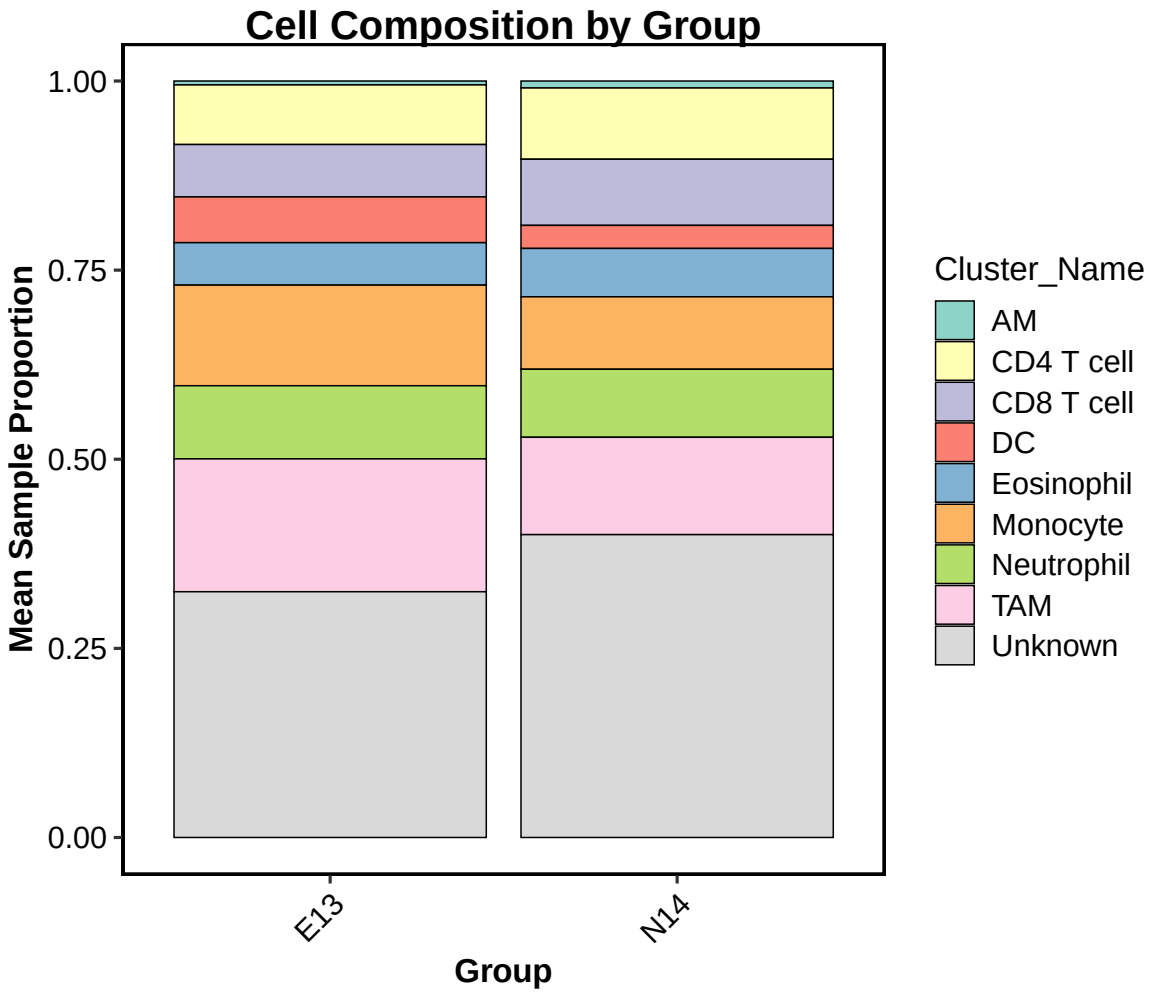
To include figures without rerunning the analysis, embed the already generated PDF or PNG files explicitly with `knitr::include_graphics()`. This chunk is intentionally set to `eval = TRUE` because it only reads existing image files; it does not rerun preprocessing, clustering, annotation, projection, or statistics.

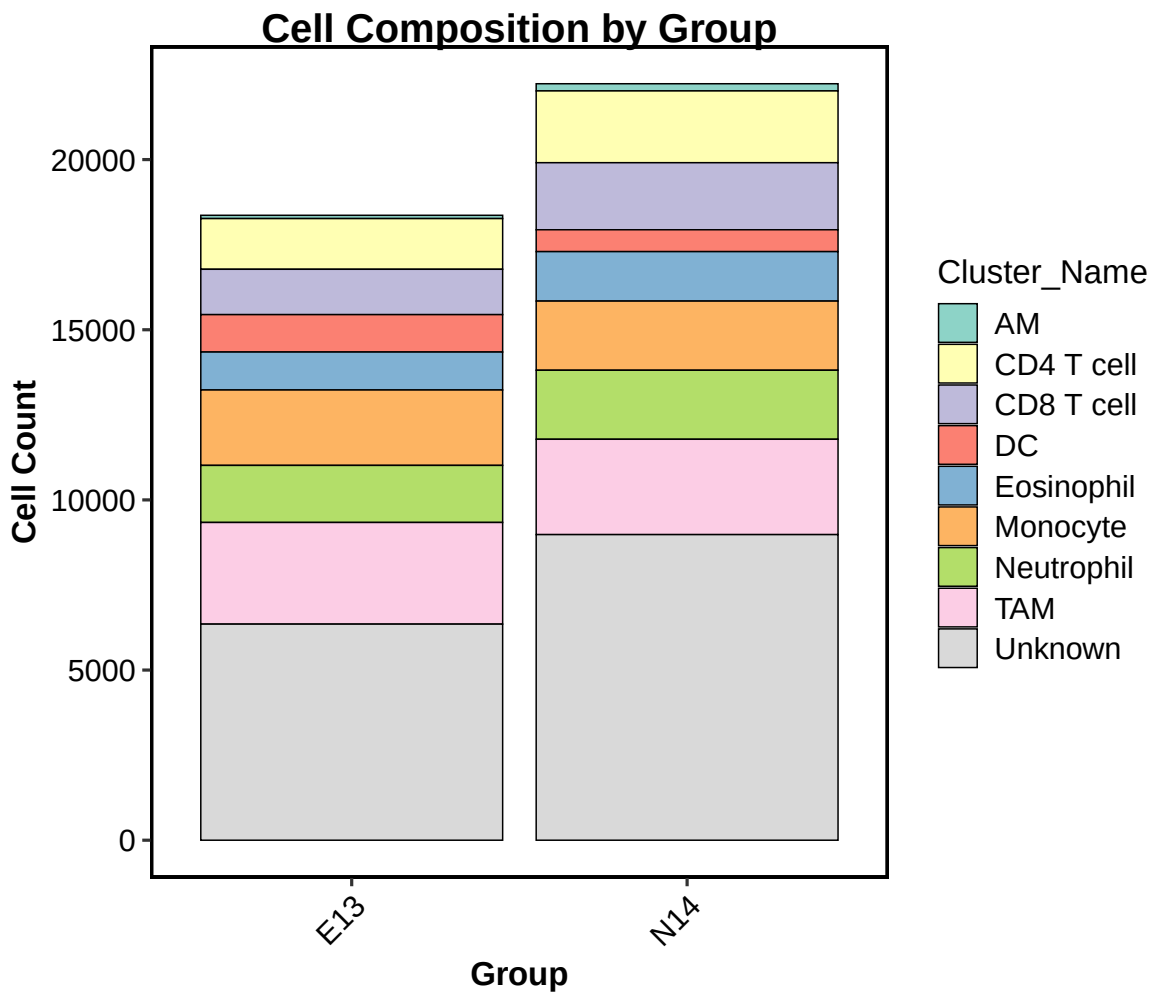
Cell Gating Hierarchy Tree

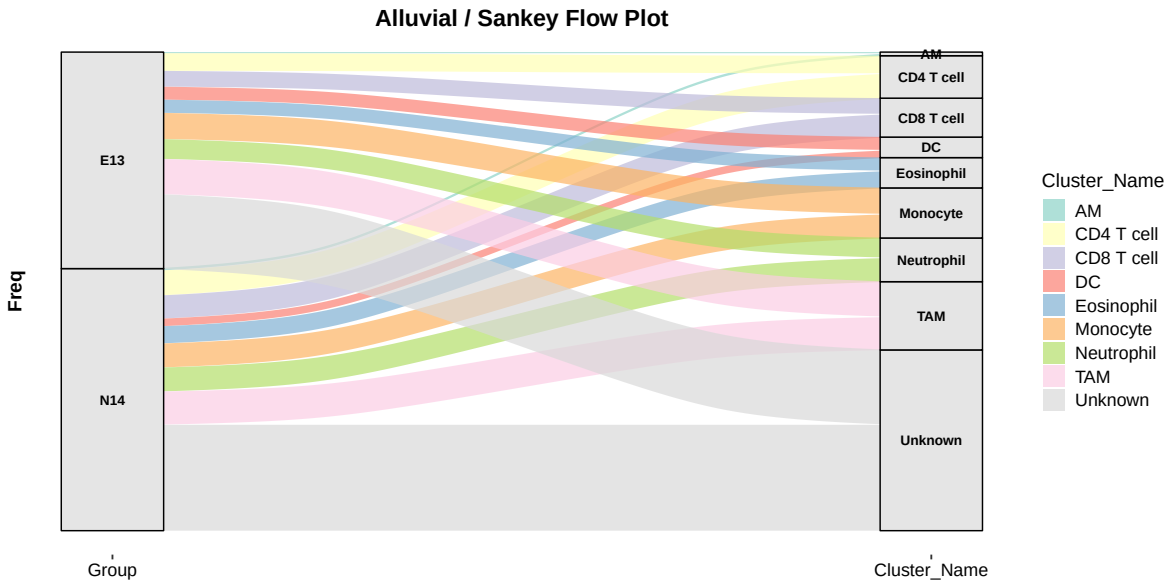


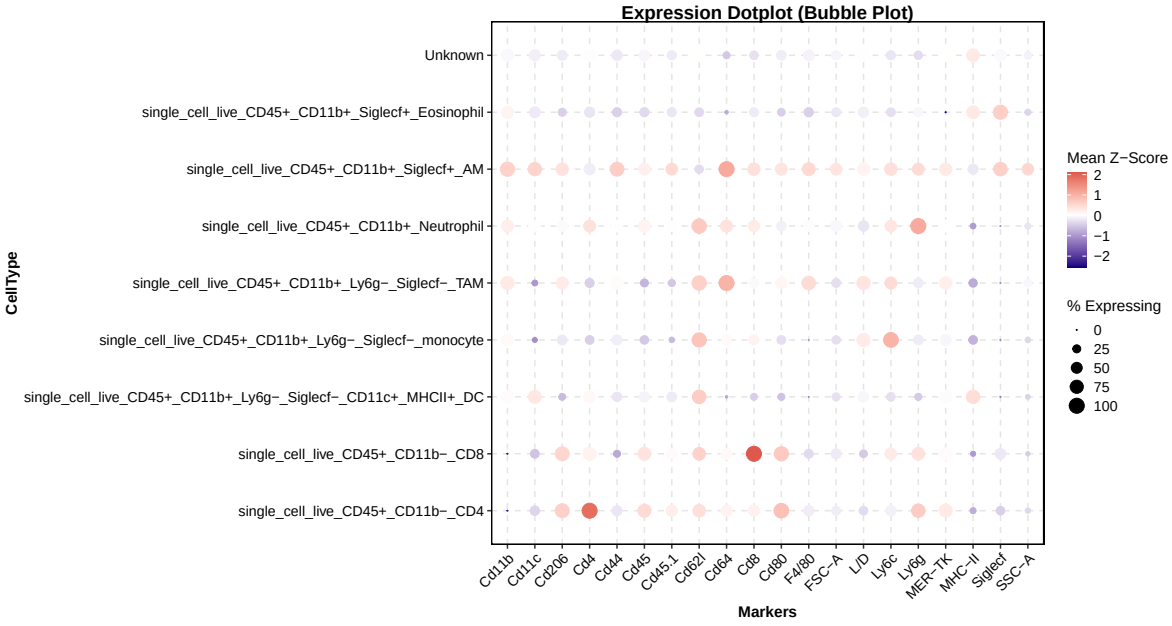


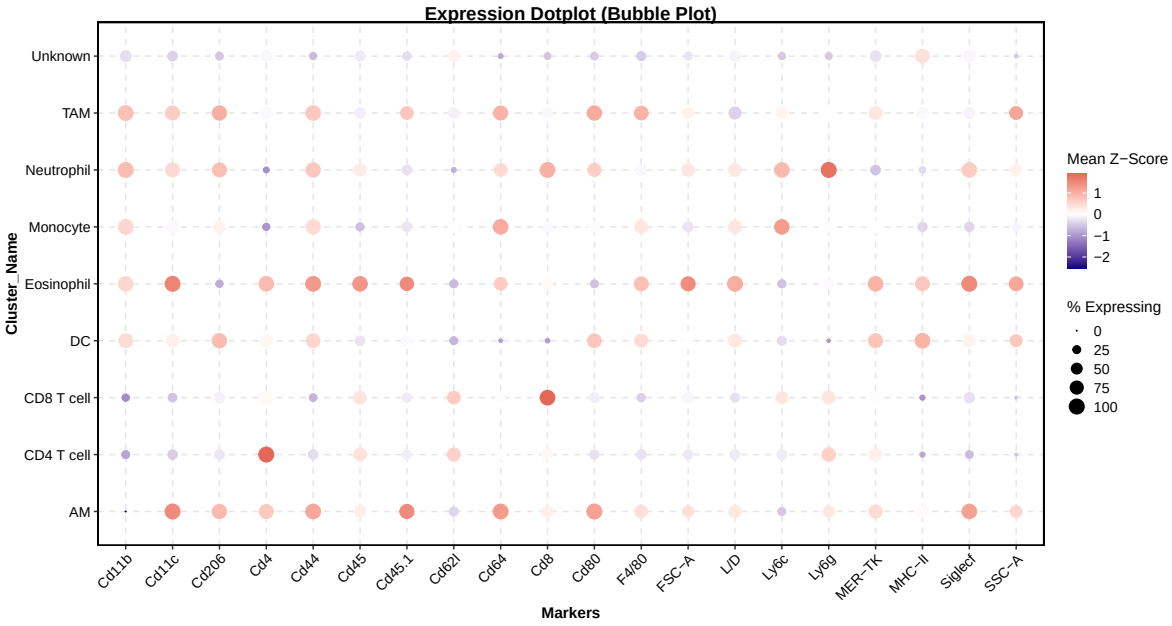

UMAP Map colored by Cluster_Name
E13
N14
UMAP2
UMAP1
AM
CD4 T cell
CD8 T cell
Eosinophil
Monocyte
Neutrophil
TAM
Unknown

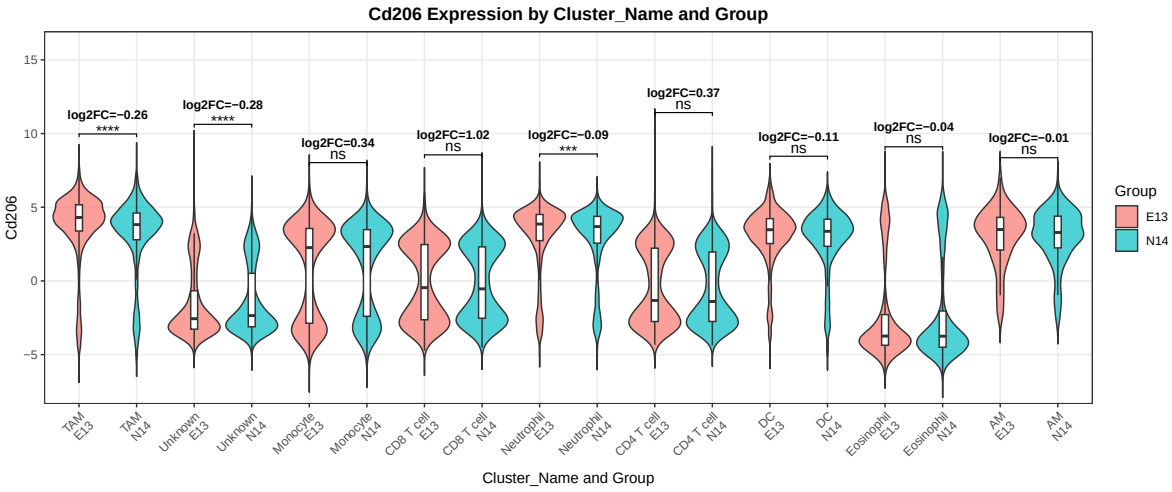

Cell Composition by Group
Mean Sample Proportion
1.00
0.75
0.50
0.25
0.00
E13
N14
Group
Cluster_Name
AM
CD4 T cell
CD8 T cell
DC
Eosinophil
Monocyte
Neutrophil
TAM
Unknown

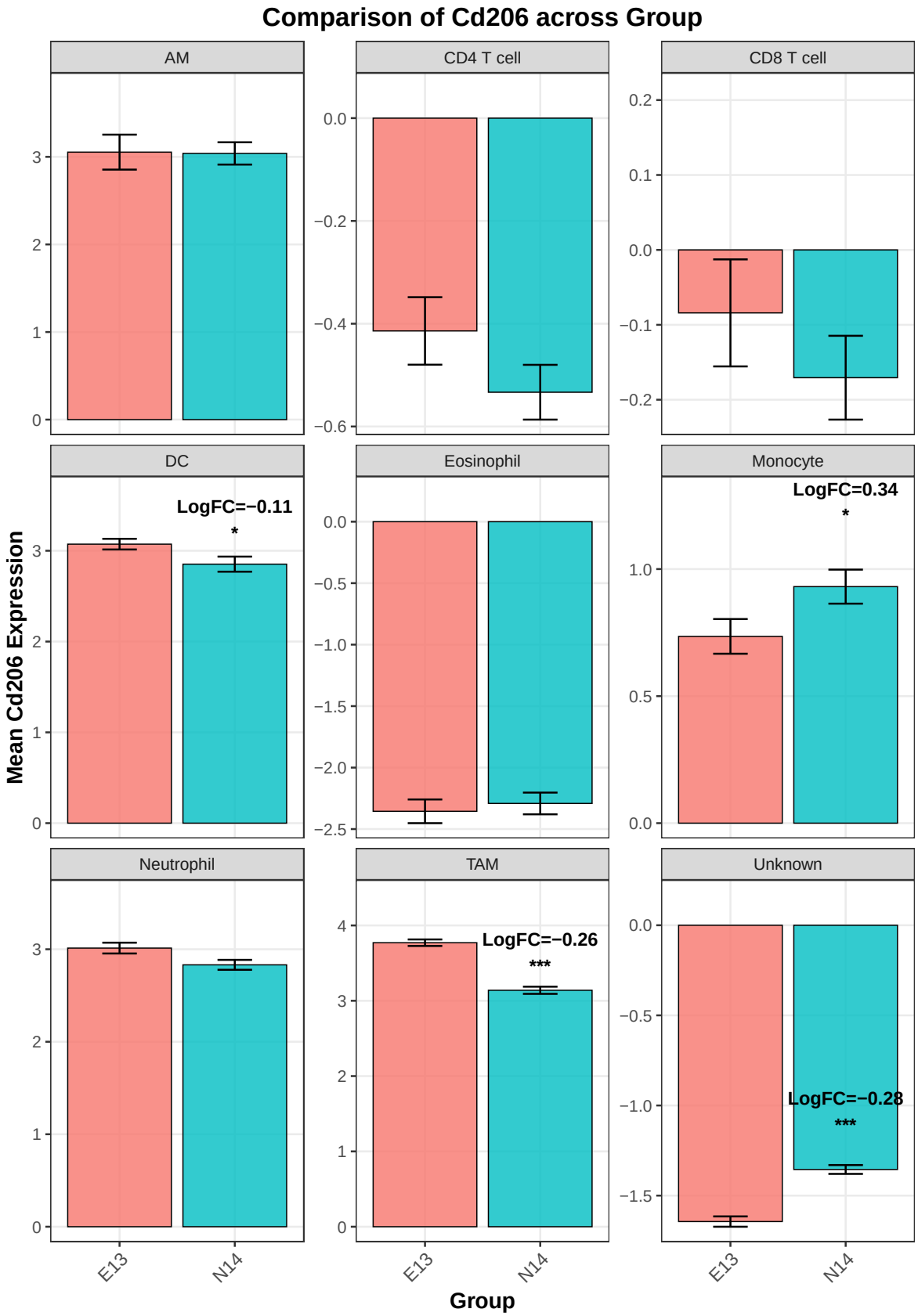

Cell Composition by Group
Cell Count
20000
15000
10000
5000
0
E13
N14
Group
Cluster_Name
AM
CD4 T cell
CD8 T cell
DC
Eosinophil
Monocyte
Neutrophil
TAM
Unknown

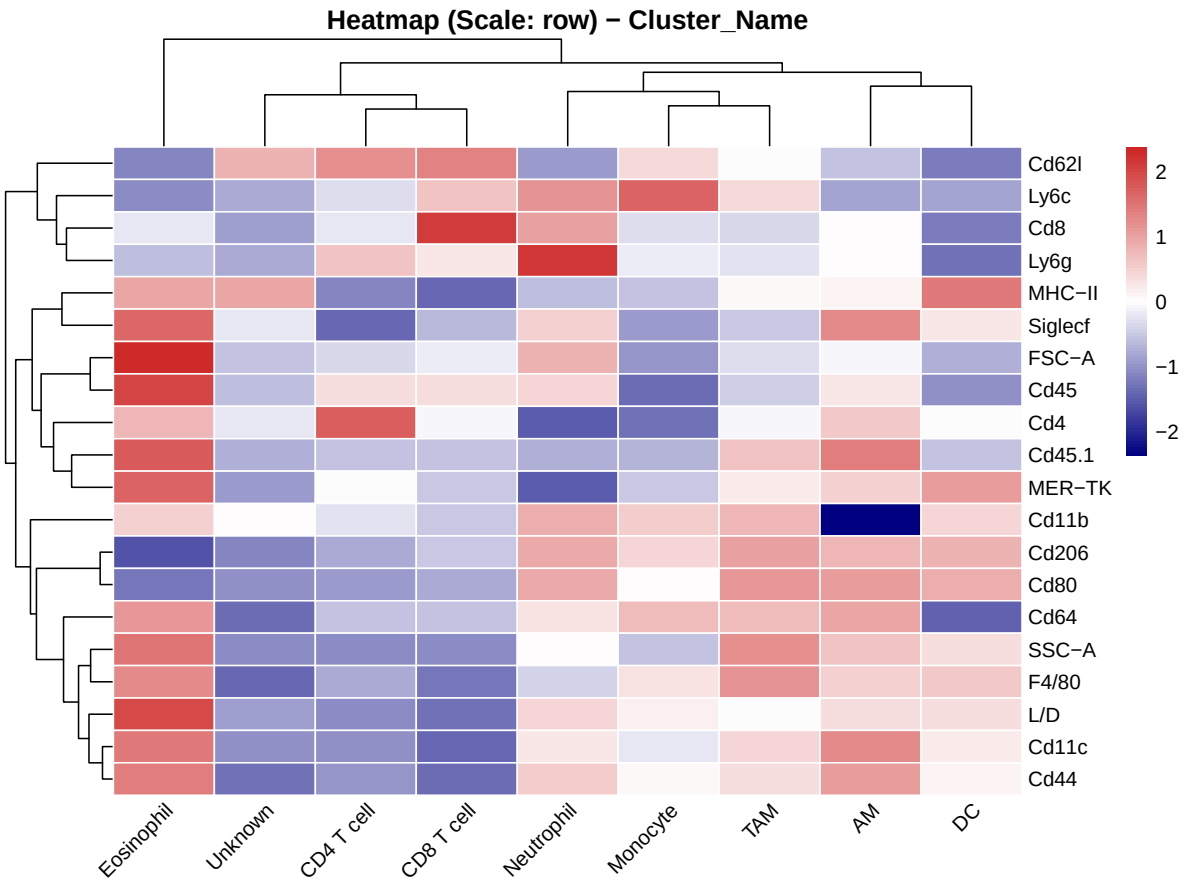

Alluvial / Sankey Flow Plot
Freq
E13
N14
Group
AM
CD4 T cell
CD8 T cell
DC
Eosinophil
Monocyte
Neutrophil
TAM
Unknown
Cluster_Name
Cluster_Name
AM
CD4 T cell
CD8 T cell
DC
Eosinophil
Monocyte
Neutrophil
TAM
Unknown

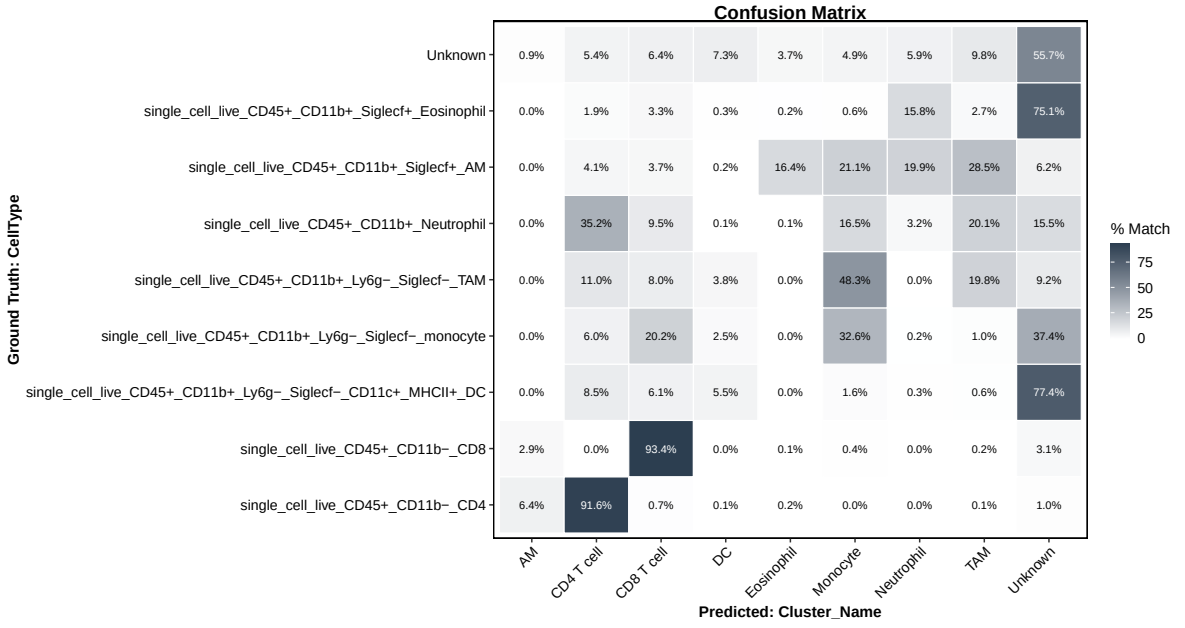

Expression Dotplot (Bubble Plot)
CellType
Unknown
single_cell_live_CD45+_CD11b+_Siglecf+_Eosinophil
single_cell_live_CD45+_CD11b+_Siglecf+_AM
single_cell_live_CD45+_CD11b+_Neutrophil
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_TAM
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_monocyte
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_CD11c+_MHCII+_DC
single_cell_live_CD45+_CD11b−_CD8
single_cell_live_CD45+_CD11b−_CD4
Cd11b
Cd11c
Cd206
Cd4
Cd44
Cd45
Cd45.1
Cd62l
Cd64
Cd8
Cd80
F4/80
FSC−A
L/D
Ly6c
Ly6g
MER−TK
MHC−II
Siglecf
SSC−A
Markers
Mean Z−Score
2
1
0
−1
−2
% Expressing
0
25
50
75
100

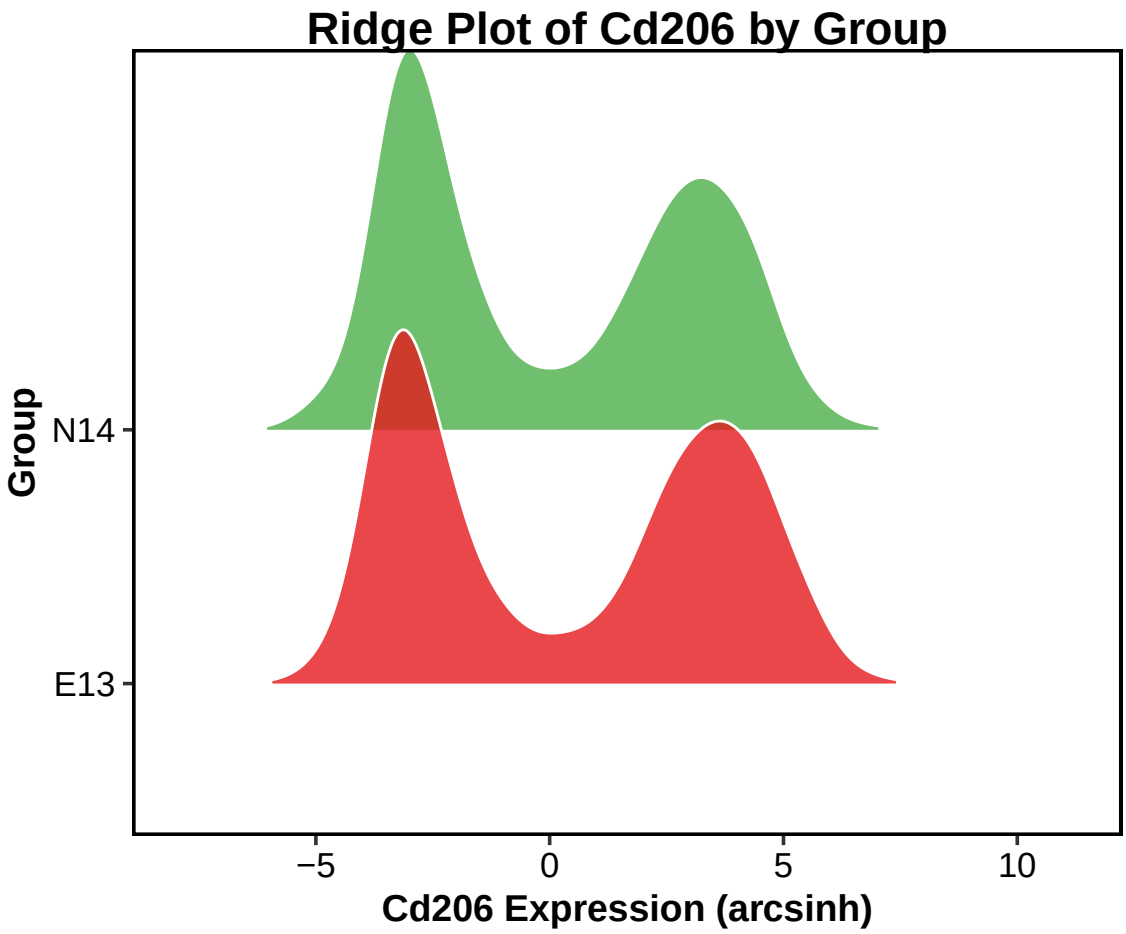

Expression Dotplot (Bubble Plot)
Cluster_Name
Unknown
TAM
Neutrophil
Monocyte
Eosinophil
DC
CD8 T cell
CD4 T cell
AM
Cd11b
Cd11c
Cd206
Cd4
Cd44
Cd45
Cd45.1
Cd62l
Cd64
Cd8
Cd80
F4/80
FSC−A
L/D
Ly6c
Ly6g
MER−TK
MHC−II
Siglecf
SSC−A
Markers
Mean Z−Score
1
0
−1
−2
% Expressing
0
25
50
75
100

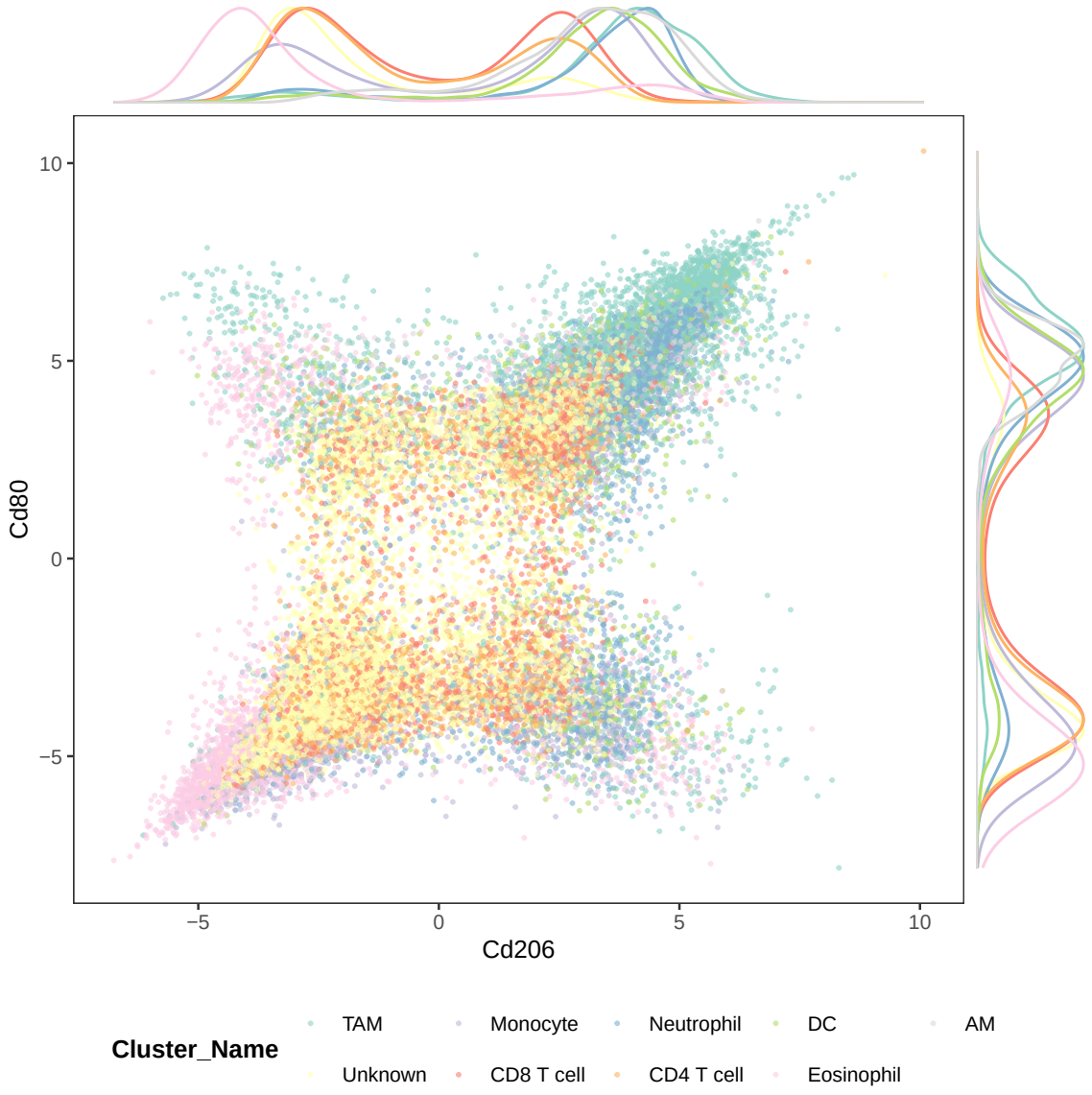

Cd206 Expression by Cluster_Name and Group
Cd206
15
10
5
0
−5
log2FC=−0.26
****
log2FC=−0.28
****
log2FC=0.34
ns
log2FC=1.02
ns
log2FC=−0.09
***
log2FC=0.37
ns
log2FC=−0.11
ns
log2FC=−0.04
ns
log2FC=−0.01
ns
TAM E13
TAM N14
Unknown E13
Unknown N14
Monocyte E13
Monocyte N14
CD8 T cell E13
CD8 T cell N14
Neutrophil E13
Neutrophil N14
CD4 T cell E13
CD4 T cell N14
DC E13
DC N14
Eosinophil E13
Eosinophil N14
AM E13
AM N14
Cluster_Name and Group
Group
E13
N14

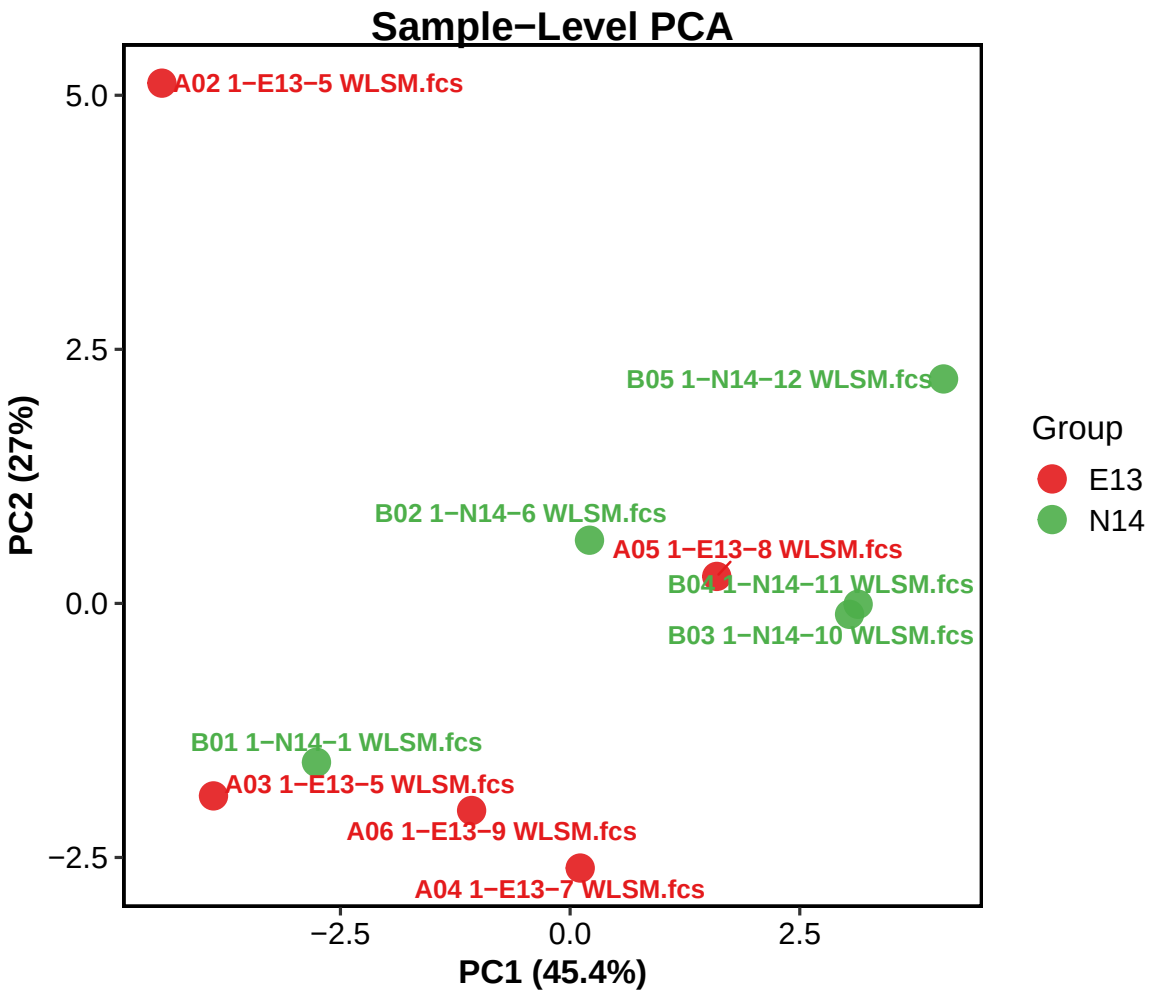

Comparison of Cd206 across Group
AM
CD4 T cell
CD8 T cell
DC
LogFC=−0.11
*
Eosinophil
Monocyte
LogFC=0.34
*
Neutrophil
TAM
LogFC=−0.26
***
Unknown
LogFC=−0.28
***
Mean Cd206 Expression
E13
N14
Group


Heatmap (Scale: row) − Cluster_Name
Cd62l
Ly6c
Cd8
Ly6g
MHC−II
Siglecf
FSC−A
Cd45
Cd4
Cd45.1
MER−TK
Cd11b
Cd206
Cd80
Cd64
SSC−A
F4/80
L/D
Cd11c
Cd44
2
1
0
−1
−2
Eosinophil
Unknown
CD4 T cell
CD8 T cell
Neutrophil
Monocyte
TAM
AM
DC


Confusion Matrix
Ground Truth: CellType
Predicted: Cluster_Name
% Match
75
50
25
0
Unknown
single_cell_live_CD45+_CD11b+_Siglecf+_Eosinophil
single_cell_live_CD45+_CD11b+_Siglecf+_AM
single_cell_live_CD45+_CD11b+_Neutrophil
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_TAM
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_monocyte
single_cell_live_CD45+_CD11b+_Ly6g−_Siglecf−_CD11c+_MHCII+_DC
single_cell_live_CD45+_CD11b−_CD8
single_cell_live_CD45+_CD11b−_CD4
AM
CD4 T cell
CD8 T cell
DC
Eosinophil
Monocyte
Neutrophil
TAM
Unknown


Ridge Plot of Cd206 by Group
Group
N14
E13
−5
0
5
10
Cd206 Expression (arcsinh)


10
5
0
−5
Cd80
−5
0
5
10
Cd206
Cluster_Name
TAM
Monocyte
Neutrophil
DC
AM
Unknown
CD8 T cell
CD4 T cell
Eosinophil



5.7.1 Adjustable parameters in the static figure-embedding chunk

Parameter	Meaning	Practical guidance
`static_figure_dir`	folder containing already generated figures	edit this to the real output folder before exporting the PDF
`static_figure_files`	ordered list of figure filenames to embed	remove figures you do not want, or add newly generated figures
`eval = TRUE`	allows only this lightweight embedding chunk to run	safe because it only reads image files and does not rerun the analysis
`out.width`	displayed width of each embedded figure	use `"95%"` for full-page figures, or smaller values for compact manuals

5.8 Optional Large-Scale Machine-Learning Projection

This section has intentionally been placed after the publication-ready figure section, as requested. It remains an optional extension of the core workflow and is most useful when a high-quality reference atlas has already been finalized and you want to project labels onto very large datasets.

5.8.1 Why ADTnorm-lite is included here

The current `05_MachineLearning.R` includes an ADTnorm-inspired correction option named `adtnorm_lite`. It is designed for large-scale projection across multiple flow/CITE-like protein batches. Before the random-forest projection step, ADTnorm-lite aligns marker-wise expression distributions across batches by matching density-derived landmarks. If density peaks and valleys cannot be matched reliably, the function falls back to robust quantile landmark alignment.

In practical terms, this option is useful when the same panel was acquired in more than one experimental batch and direct projection produces visible batch separation. The correction is applied only to the numeric marker features used by the model; metadata columns such as `Batch`, `Group`, `SampleID`, `Cluster_Name`, and UMAP coordinates are ignored.

Important requirement: the updated `05_MachineLearning.R` must be sourced before running this section. If R reports `unused arguments (batch_correction = ..., batch_col = ...)`, the session is still using an older `05_MachineLearning.R`.

5.8.2 Prepare two projection batches

The following block mirrors the current `EasyFlow Analysis.R` section titled `# Step 4b (optional): large-scale machine-learning projection`. The two folders represent two acquisition batches. Each batch is preprocessed and panel-mapped separately, then a `Batch` column is added before merging.

```
batch1_folder <- projection_cfg$batch1_folder
batch2_folder <- projection_cfg$batch2_folder

batch1_label <- projection_cfg$batch1_label
batch2_label <- projection_cfg$batch2_label
gating_template_path <- projection_cfg$gating_template_path

batch1_raw_data <- Prepare_Raw_Data(
  work_dir = batch1_folder,
  group_keywords = group_keywords,
  n_cells_per_sample = projection_cfg$n_cells_per_sample,
  do_QC = preprocess_cfg$do_QC,
  do_compensation = preprocess_cfg$do_compensation,
  transform_method = preprocess_cfg$transform_method,
  cofactor = preprocess_cfg$cofactor
)
batch1_raw_data <- Apply_Panel_Mapping(batch1_raw_data, my_panel)
batch1_raw_data$Batch <- batch1_label
```

```
batch2_raw_data <- Prepare_Raw_Data(
  work_dir = batch2_folder,
  group_keywords = group_keywords,
  n_cells_per_sample = projection_cfg$n_cells_per_sample,
  do_QC = preprocess_cfg$do_QC,
  do_compensation = preprocess_cfg$do_compensation,
  transform_method = preprocess_cfg$transform_method,
  cofactor = preprocess_cfg$cofactor
)
batch2_raw_data <- Apply_Panel_Mapping(batch2_raw_data, my_panel)
batch2_raw_data$Batch <- batch2_label
```

5.8.3 Adjustable parameters for two-batch preparation

Parameter	Current example value	Meaning	Practical guidance
`batch1_folder`	`"/Volumes/1 TB/lly go colitis facs/20240619 colitis cytokine/mln"`	folder containing the first batch of FCS files	edit for the reference or baseline acquisition batch
`batch2_folder`	`"/Volumes/1 TB/lly go colitis facs/situmulate mlns"`	folder containing the second batch of FCS files	edit for the additional acquisition batch
`batch1_label`	`"Batch1_20240619_mLN"`	label written to `Batch` for batch 1	use a short, stable, publication-friendly label
`batch2_label`	`"Batch2_stimulated_mLN"`	label written to `Batch` for batch 2	use the same naming style as `batch1_label`
`group_keywords`	`c("wt", "ko")`	filename keywords used to infer biological groups	edit when the group names differ
`n_cells_per_sample`	`1500000`	maximum number of cells sampled per FCS file for projection	reduce for testing; increase for final large-scale projection
`Batch`	created manually	metadata column identifying acquisition batch	required when `batch_correction = "adtnorm_lite"`

5.8.4 Recommended template-gating call

```
batch1_raw_data <- Apply_Gating_Template(
  df = batch1_raw_data,
```

```
  target_node = projection_cfg$target_node,
  template_path = gating_template_path,
  margin = projection_cfg$margin
)

batch2_raw_data <- Apply_Gating_Template(
  df = batch2_raw_data,
  target_node = projection_cfg$target_node,
  template_path = gating_template_path,
  margin = projection_cfg$margin
)

massive_raw_data <- dplyr::bind_rows(batch1_raw_data, batch2_raw_data)
massive_raw_data$Batch <- factor(
  massive_raw_data$Batch,
  levels = c(batch1_label, batch2_label)
)
```

5.8.5 Adjustable parameters in `Apply_Gating_Template()`

Parameter	Current example value	Meaning	Practical guidance
`df`	`batch1_raw_data` or `batch2_raw_data`	large dataset to be filtered	apply the same template independently to each batch before merging
`target_node`	`"cell_live_cell_CD4"`	gating-template node used as the global filter	must exactly match the node name exported from the gating template
`template_path`	`projection_cfg$gating_template_path`	path to the gating template	use an absolute path when knitting from another working directory
`margin`	`0.02`	shrinks the template boundaries by this amount	increase for stricter filtering, but too much may remove true cells

5.8.6 Main large-scale projection call with ADTnorm-lite

```
final_massive_results <- Project_Atlas_To_Massive_Data(
  ref_obj = final_results,
  massive_data = massive_raw_data,
  target_label = projection_cfg$target_label,
```

```
  cv_folds = projection_cfg$cv_folds,
  rejection_threshold = projection_cfg$rejection_threshold,
  use_class_weights = projection_cfg$use_class_weights,
  rare_cell_boost = projection_cfg$rare_cell_boost,
  batch_correction = projection_cfg$batch_correction,
  batch_col = projection_cfg$batch_col,
  reference_batch = batch1_label,
  batch_diagnostic_pdf = NULL
)

cat(
  "Large-scale projection completed:",
  nrow(final_massive_results$Data), "cells projected.\n"
)
```

Expected ADTnorm-lite outputs:

- `ADTnorm_Lite_Batch_Correction_Diagnostics.csv`: marker-by-marker correction summary, written to the current working directory by default.
- `ADTnorm_Batch_Correction_Diagnostics.pdf`: before/after density plots for each corrected marker, written to the current working directory in the example above.
- `final_massive_results$Info$Batch_Correction`: stored diagnostic information when available.

5.8.7 Optional plotting of projected results

```
final_massive_results <- Set_Plot_Order(
  final_massive_results,
  order_list = list(
    Group = c("wt", "ko"),
    Batch = c(batch1_label, batch2_label)
  )
)

projection_dim <- "UMAP"
projection_color <- "Cluster_Name"
projection_split <- "Batch"

p_dim_projected <- Plot_Dim_Reduction(
  final_massive_results,
  dim_method = projection_dim,
  color_by = projection_color,
  split_by = projection_split
)
```

```
out_dim_name <- paste0(
  "01_", projection_dim, "_by_", projection_color,
  "_split_", projection_split,
  "_ADTnormLite.pdf"
)

p_dim_projected
ggsave(out_dim_name, plot = p_dim_projected, width = 12, height = 6)
```

5.8.8 Adjustable parameters in `Project_Atlas_To_Massive_Data()`

Parameter	Current example value	Meaning	Practical guidance
`ref_obj`	`final_results`	annotated reference atlas	should be a high-quality, well-annotated reference object
`massive_data`	`massive_raw_data`	large target dataset to predict	fixed output from the previous step
`target_label`	`"Cluster_Name"`	label column to learn and project	can be changed to `CellType` or another valid label column
`cv_folds`	`5`	number of CV folds	set to `1` if you want to skip cross-validation
`rejection_threshold`	`0.55`	probability threshold below which cells become `Unassigned`	increase for stricter rejection, decrease for more aggressive assignment
`use_class_weights`	`TRUE`	enables rare-class weighting	recommended when rare populations matter biologically
`rare_cell_boost`	`2.0`	strength of rare-class protection	larger values protect rare populations more strongly but may also increase misassignment risk
`batch_correction`	`"adtnorm_lite"`	enables marker-wise batch alignment before projection	set to `"none"` for single-batch data or when no batch correction is desired
`batch_col`	`"Batch"`	metadata column identifying the acquisition batch	must exist in `massive_data` before projection

Parameter	Current example value	Meaning	Practical guidance
`reference_batch`	`"Batch1_20240619_mLN"`	batch used as the expression-scale reference	usually choose the cleaner, baseline, or reference acquisition batch
`batch_diagnostic_pdf`	`"ADTnorm_Batch_Correction_Diagnostics.pdf"`	output PDF with before/after density diagnostics	inspect this file to confirm that correction is biologically reasonable

5.8.9 How to interpret ADTnorm-lite diagnostics

The diagnostic PDF should show each marker's density distribution before and after correction. Good correction usually reduces broad batch shifts while preserving biologically meaningful positive and negative populations. If the after-correction densities look over-flattened or biologically implausible, rerun the projection with:

```
projection_cfg$batch_correction <- "none"
```

and compare UMAPs, dotplots, and sample-level abundance patterns between the corrected and uncorrected results.

5.8.10 Rapid projection for future new samples

```
if (exists("new_sample_data")) {
  new_sample_result <- Project_New_Sample(
    ref_obj = final_massive_results,
    new_data = new_sample_data,
    rejection_threshold = projection_cfg$rejection_threshold
  )
} else {
  message("new_sample_data was not found; skipping rapid new-sample projection.")
}
```

5.8.11 Adjustable parameters in `Project_New_Sample()`

Parameter	Current example value	Meaning
`ref_obj`	`final_massive_results`	projection object containing the trained RF model
`new_data`	`new_sample_data`	new sample-level data frame
`target_label`	`"Cluster_Name"`	output label column

Parameter	Current example value	Meaning
`rejection_threshold`	`0.55`	low-confidence rejection threshold

5.9 Step 7: Statistical Analysis And Report Output

5.9.1 Differential abundance statistics

```
stats_results_rds <- file.path(output_dir, "stats_results.rds")

if (file.exists(stats_results_rds)) {
  stats_results <- readRDS(stats_results_rds)
  message("Loaded stats_results from: ", stats_results_rds)
} else {
  stats_results <- Perform_Abundance_Stats(
    res_obj = final_results,
    target_label = stats_cfg$my_stat_target,
    group_col = stats_cfg$my_stat_group,
    method = stats_cfg$my_stat_method,
    transform = stats_cfg$my_stat_trans
  )
  if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)
  saveRDS(stats_results, file = stats_results_rds)
  message("Saved stats_results to: ", stats_results_rds)
}

if (!is.null(stats_results$Stats) && nrow(stats_results$Stats) > 0) {
  knitr::kable(stats_results$Stats, caption = "Abundance statistics results")
} else if (is.data.frame(stats_results)) {
  knitr::kable(stats_results, caption = "Abundance statistics results")
} else {
  print(stats_results)
}
cat("Abundance testing completed:", nrow(stats_results$Stats), "rows in the statistics table.\
```

5.9.1.1 Adjustable parameters in `Perform_Abundance_Stats()`

Parameter	Current example value	Meaning	Practical guidance
`res_obj`	`final_results`	final result object	fixed output from the previous step
`target_label`	`"Cluster_Name"`	label used to define populations	`Cluster_Name` is the most common choice

Parameter	Current example value	Meaning	Practical guidance
`group_col`	`"Group"`	sample-level grouping variable	must be a valid sample-level metadata column
`method`	`"wilcoxon"`	one of `"wilcoxon"`, `"t.test"`, `"kruskal"`, `"anova"`, or `"lmm"`	Wilcoxon or t-test are common for two-group comparisons
`transform`	`"clr"`	one of `"none"`, `"asin_sqrt"`, or `"clr"`	`clr` is often preferred for compositional abundance data
`random_effect`	`NULL`	random-effect column for mixed models	only used when `method = "lmm"`
`sample_col`	`"SampleID"`	sample ID column	usually does not need to be changed

5.9.2 Optional marker differential expression analysis

```
de_results_rds <- file.path(output_dir, "de_results.rds")

if (file.exists(de_results_rds)) {
  de_results <- readRDS(de_results_rds)
  message("Loaded de_results from: ", de_results_rds)
} else {
  de_results <- tryCatch(
    Perform_Marker_DE(
      res_obj = final_results,
      group_col = stats_cfg$my_stat_group,
      cluster_col = "Cluster_Name",
      method = "wilcoxon"
    ),
    error = function(e) { message("Marker DE failed: ", e$message); NULL }
  )
  if (!is.null(de_results)) {
    saveRDS(de_results, file = de_results_rds)
    message("Saved de_results to: ", de_results_rds)
  }
}

if (!is.null(de_results) && nrow(de_results) > 0) {
  knitr::kable(head(de_results, 30), caption = "Marker differential expression results (top 30)
  cat("Marker differential analysis completed:", nrow(de_results), "rows exported.\n")
}
```

5.9.2.1 Adjustable parameters in `Perform_Marker_DE()`

Parameter	Default	Meaning
`cluster_name`	`NULL`	analyze one cluster only, or all clusters when `NULL`
`group_col`	`"Group"`	grouping variable used for the comparison
`cluster_col`	`"Cluster_Name"`	cluster label column
`markers`	`NULL`	custom marker list; `NULL` triggers automatic detection
`method`	`"wilcoxon"` / `"t.test"`	two-group marker comparison method

5.9.3 Trajectory inference

```
traj_res_rds <- file.path(output_dir, "traj_res.rds")

if (file.exists(traj_res_rds)) {
  traj_res <- readRDS(traj_res_rds)
  final_results <- traj_res$res_obj
  message("Loaded traj_res from: ", traj_res_rds)
} else {
  traj_res <- tryCatch(
    Infer_Trajectory(
      res_obj = final_results,
      start_cluster = NULL,
      use_dim = "UMAP",
      out_dir = trajectory_dir
    ),
    error = function(e) { message("Trajectory inference failed: ", e$message); NULL }
  )
  if (!is.null(traj_res)) {
    final_results <- traj_res$res_obj
    saveRDS(traj_res, file = traj_res_rds)
    message("Saved traj_res to: ", traj_res_rds)
  }
}

if (!is.null(traj_res)) {
  cat(
    "Trajectory inference completed. Output files:",
    paste(unlist(traj_res$plot_files), collapse = ", "),
    "\n"
  )
}
```

```r
if (exists("traj_res") && !is.null(traj_res)) {
  trajectory_files <- unlist(traj_res$plot_files)
  trajectory_files <- trajectory_files[file.exists(trajectory_files)]
  if (length(trajectory_files) > 0) {
    knitr::include_graphics(trajectory_files)
  } else {
    cat("Trajectory figures were not found in:", trajectory_dir, "\n")
  }
}
```

5.9.3.1 Adjustable parameters in `Infer_Trajectory()`

Parameter	Default	Meaning
`start_cluster`	`NULL`	explicitly defines the starting cluster
`start_by_col`	`NULL`	infers the starting state from an annotation column such as `CellType`
`start_by_value`	`NULL`	the annotation value used together with `start_by_col`
`start_mode`	`"majority"` or `"centroid"`	how the starting state is mapped to a cluster when needed
`use_dim`	`"UMAP"`	embedding used for trajectory inference
`out_dir`	`"09_Trajectory_Outputs"`	output directory for trajectory figures
`prefix`	`"Trajectory"`	filename prefix
`point_size`	`0.5`	point size in the exported trajectory plots
`seed`	`123`	random seed

5.9.4 FCS export

```r
Export_To_FCS(final_results, out_dir = "09_Final_FCS_Output")
```

5.9.4.1 Adjustable parameters in `Export_To_FCS()`

Parameter	Default	Meaning
`out_dir`	`"Annotated_FCS_Outputs"`	directory for exported annotated FCS files
`sample_col`	`"SampleID"`	sample ID column

5.9.5 HTML report generation

```
Generate_Final_HTML_Report(
  res_obj = final_results,
  stats_res = stats_results,
  project_name = report_cfg$project_name,
  out_file = report_cfg$out_file,
  template = report_cfg$template
)

Generate_Final_HTML_Report(
  res_obj = final_results,
  stats_res = stats_results,
  out_file = "EasyFlow2_Review_Report.html",
  template = "review"
)

Generate_Final_HTML_Report(
  res_obj = final_results,
  stats_res = stats_results,
  out_file = "EasyFlow2_Custom_Report.html",
  include_plots = c(
    "sample_counts", "embedding", "composition",
    "marker_dotplot", "trajectory", "abundance"
  )
)

saveRDS(final_results, file.path(output_dir, "final_results.rds"))
cat("Final result object saved to:", file.path(output_dir, "final_results.rds"), "\n")
```

5.9.5.1 Adjustable parameters in `Generate_Final_HTML_Report()`

Parameter	Default	Meaning
`res_obj`	`final_results`	final result object
`stats_res`	`stats_results`	abundance statistics result
`project_name`	`"EasyFlow2 Final Report"`	report title
`out_file`	`"EasyFlow2_Final_Report.html"`	HTML output filename
`sample_col`	`"SampleID"`	sample column name
`group_col`	`"Group"`	group column name
`template`	`"manuscript"`	one of `"manuscript"`, `"review"`, or `"full"`
`include_plots`	`NULL`	custom selection of plots to include
`top_markers_n`	`10`	number of highly variable markers shown in the report

Parameter	Default	Meaning
`top_stats_n`	`15`	number of top statistical results shown in the report
`true_label`	`"CellType"`	true-label column used in the confusion matrix
`pred_label`	`"Cluster_Name"`	predicted-label column used in the confusion matrix

5.9.5.2 Allowed `include_plots` values

Value	Meaning
`sample_counts`	retained cell counts per sample
`embedding`	UMAP/t-SNE overview
`composition`	group-level composition
`marker_dotplot`	marker identity dotplot
`sample_pca`	sample-level PCA
`trajectory`	trajectory and pseudotime section
`abundance`	abundance statistics section
`confusion`	confusion matrix section
`metadata`	pipeline metadata section

6 Typical Output Files Generated By The Current Workflow

Stage	File or directory	Content
gating	`Gating_Thresholds_Template.csv`	gating threshold template
gating	`Gating_Tree_Structure.csv`	gating tree structure
gating	`00_Gating_Tree_Plot.pdf`	gating tree PDF
annotation	`Cluster_Annotation_Report.csv`	cluster annotation report
figures	`01_*.pdf` to `10_*.pdf`	publication-ready figures
statistics	`08_Differential_Abundance_Stats_*.csv`	differential abundance results
marker DE	`DE_Markers_*.csv`	marker differential-expression results
trajectory	`09_Trajectory_Outputs/`	pseudotime output figures
FCS	`09_Final_FCS_Output/`	annotated FCS exports
report	`EasyFlow2_Final_Report.html` and related files	automated HTML reports

7 Suggested Methods Text For A Manuscript

The following paragraph can be adapted to your specific biological system:

Raw FCS files were processed using a custom R-based workflow, EasyFlow2, which integrates preprocessing, interactive hierarchical gating, unsupervised clustering, annotation, machine-learning projection, and downstream reporting. During preprocessing, FCS files were imported without instrument-level transformation, followed by optional quality control, fluorescence compensation, nonlinear transformation, and per-sample downsampling. Quality control included low-end FSC-A filtering to remove debris and FSC-A/FSC-H-based singlet selection. Compensation matrices were automatically extracted from the SPILL or spillover fields of the FCS header, and fluorescence channels were transformed using either arcsinh or logicle transformation. Preprocessed single-cell matrices were then subjected to custom interactive hierarchical gating, in which users iteratively defined cell populations in arbitrary two-dimensional marker space. Parent-child relationships among gated populations were explicitly recorded as a gating tree, and percentile-based thresholds for each node were exported as reusable gating templates. Downstream unsupervised analysis included graph-based or FlowSOM clustering together with t-SNE and UMAP embedding. Cluster identities were further assigned using a dual-track annotation framework that combined rule-based marker logic with LLM-assisted semantic annotation under ontology and lineage-context constraints. For large-scale cross-batch projection, an annotated reference atlas was used to train a random-forest classifier with optional rare-class weighting and out-of-distribution rejection. When multiple acquisition batches were projected together, marker-wise batch effects were optionally reduced using an ADTnorm-inspired landmark-alignment procedure (`adtnorm_lite`) before label projection and UMAP coordinate transfer. The final outputs were used for population abundance testing, marker-level differential analysis, trajectory inference, annotated FCS export, and automated HTML report generation.

8 Practical Notes And High-Frequency Pitfalls

8.1 `function_dir` and `setwd()` are environment-specific

The absolute paths in the current project script are machine-specific. In a new project, the first items to update are usually:

- `setwd()`
- `function_dir`
- `my_folder` or `data_dir`

8.2 `Plot_Scatterhist_Line()` requires additional packages

If the dual-marker marginal-density scatter plot fails, first check:

- `patchwork`
- `cowplot`

These packages may not have been included in older versions of the setup block.

8.3 `Perform_Marker_DE()` currently supports two-group comparisons only

If `group_col` contains more than two groups, the current implementation will stop. For multi-group studies, either split the comparison into pairwise analyses or use another statistical framework.

8.4 Panel consistency is essential before large-scale projection

`Project_Atlas_To_Massive_Data()` and `Project_New_Sample()` both enforce strict feature matching between the reference atlas and new data. If a panel mismatch error appears, check:

1. whether `Apply_Panel_Mapping()` has been applied;
2. whether marker naming and capitalization are perfectly consistent;
3. whether the reference atlas and new samples truly come from the same panel.

8.5 ADTnorm-lite requires a valid `Batch` column

When `batch_correction = "adtnorm_lite"`, the merged projection dataset must contain a batch column such as `Batch`, and `reference_batch` must exactly match one of its values. If R reports unused arguments for `batch_correction`, `batch_col`, or `reference_batch`, restart the R session and source the updated `05_MachineLearning.R`.

8.6 HTML rendering requires Pandoc

If `Generate_Final_HTML_Report()` or the full R Markdown rendering fails, verify that Pandoc is available in your RStudio or system environment.

9 Minimal Reproducible Example

If you only want a compact working template, start from the following minimal example:

```
project_dir <- "/path/to/project"
data_dir <- "/path/to/fcs_folder"
setwd(project_dir)

for (file in sort(list.files(project_dir, pattern = "^[0-9]{2}_.*\\.[Rr]$", full.names = TRUE))
  source(file)
}

my_panel <- c(
  "BV605-A" = "CD4",
  "BV510-A" = "CD103",
  "FITC-A" = "GM-CSF",
  "PE-Cy7-A" = "IL17a",
  "BV421-A" = "TNFa",
```

```
  "PE-A" = "IL6",
  "AF700-A" = "IL10",
  "APC-A" = "IFNg",
  "eFluor450-A" = "Foxp3"
)

raw_data <- Prepare_Raw_Data(
  work_dir = data_dir,
  group_keywords = c("wt", "ko"),
  n_cells_per_sample = 50000,
  do_QC = TRUE,
  do_compensation = TRUE,
  transform_method = "arcsinh",
  cofactor = 150
)

raw_data <- Apply_Panel_Mapping(raw_data, my_panel)
gated_cells <- Run_Universal_Gating(raw_data)

ref_results <- Run_Analysis_On_Gated_Data(
  gated_data = gated_cells,
  k_neighbors = 30,
  cluster_mode = "B",
  do_scale = TRUE,
  cluster_method = "louvain",
  umap_n_neighbors = 15,
  umap_min_dist = 0.2
)

final_results <- Annotate_Clusters_Dual_Track(
  res_obj = ref_results,
  model = "openai/gpt-5.4",
  use_llm_only = TRUE,
  ontology = c("CD4 T cell", "Unknown"),
  base_population_context = "All cells were pre-gated as CD4+ T cells.",
  annotation_granularity = "fine"
)

stats_results <- Perform_Abundance_Stats(
  res_obj = final_results,
  target_label = "Cluster_Name",
  group_col = "Group",
  method = "wilcoxon",
  transform = "clr"
)

Generate_Final_HTML_Report(
```

```
  res_obj = final_results,
  stats_res = stats_results,
  out_file = "EasyFlow2_Final_Report.html",
  template = "manuscript"
)
```

10 Closing Note

The current EasyFlow codebase already supports a complete workflow from raw FCS input through interactive gating, dimensionality reduction, dual-track annotation, machine-learning projection, publication-grade visualization, statistical analysis, trajectory inference, and automated HTML reporting. In practice, the three most important aspects to maintain consistently are:

1. path and directory configuration;
2. panel naming and marker naming consistency;
3. consistent use of label columns across annotation, plotting, and statistics.

If these three components remain aligned, downstream tuning and reproducibility become much easier.

11 Session Information

```
sessionInfo()
```